UPSC Prelims

Environment & Ecology

Class No. 14

Energy Part 2 and Pollution Part 1

- **Cassava (Tapioca)**
  - **ICAR-Central Tuber Crops Research Institute (CTCRI)** has identified Cassava (tapioca) as a promising raw material for bioethanol production to meet **Ethanol Blending Petrol (EBP) programme target of 2025.**
- **About Cassava:**
  - Its starch with its unique physico-chemical and functional properties finds extensive applications in the food and industrial sectors.
  - The agricultural residues of cassava such as peels, stems and leaves are potential feedstock for **2G bioethanol production.**
  - Major production is from Tamil Nadu, followed by Kerala.

Flexi Fuel

- **Context:** Ministry of Road Transport & Highways has launched first of its kind pilot project on Flexi-Fuel Strong Hybrid Electric Vehicles (FFV-SHEV) in India.

Flex Fuel vehicles (FFV)

- Like traditional vehicles, flex fuel vehicles have an internal combustion engine, but instead of regular petrol, it can run on **blended fuel — petrol with ethanol or methanol.**
    - The ethanol mix can vary between 20% and 85%.
- Unlike electric hybrid vehicles, no bulky parts need to be added to the basic gasoline vehicle architecture.
- Upgrading existing vehicles is possible but expensive and not considered feasible.

Renewable Energy

- Flexi Fuel Vehicles are currently being **used successfully in Brazil,** giving people the option to switch fuel (gasoline and ethanol).
- An **Flexi-Fuel Strong Hybrid Electric Vehicles** (FFV-SHEV) possesses a flex-fuel engine and an electric powertrain.
  - This setup extends dual advantages of higher ethanol use and greater fuel efficiency, as it can run on its EV mode for a good amount of time, while the engine stands shut off.
- **Significance of FFV:** Less polluting, check on diversion of sugar used in ethanol blending, burn with whatever proportion of mixture is in its combustion chamber, reduced import bill etc.
- **Challenges of FFV:** lack of adequate infrastructure, issue with ethanol supply since this largely comes from sugarcane in India, **less mileage** etc.

Renewable Energy

Different types of vehicles as per fuel

Mono-fuel vehicle

Vehicle which operates on natural gas only. Also known as Dedicated Natural Gas Vehicle.

Bi-fuel vehicle

Vehicle that has two independent fuel systems (one of them for natural gas) and can run on both fuels simultaneously.

Flex fuel vehicles (FFVs)

FFV is any motor vehicle (or motor vehicle engine) engineered and designed to be operated on any mixture of two or more different fuels.

- FFVs are capable of running on any blend of gasoline and ethanol up to 83%.

Alternate Fuels - A Comparison

Fuel Type	Significance	Challenges
Ethanol as fuel	▪ **Renewable, domestically produced** transportation fuel. ▪ Has a **higher octane number** than gasoline, which provides increased power and performance. ▪ Ethanol production creates jobs in rural areas	▪ **Sporadic availability of ethanol** ▪ Ethanol can also cause **corrosion and damage to the engine.** ▪ Ethanol is also **not as economical as gasoline** as it does not provide the same level of fuel efficiency.
Flex fuel	▪ With much lower cost of running, they also offer **better economy for consumers.**	▪ The **benefit for the environment is less as compared to battery EVs** or hydrogen fuel cell vehicles of the future
Hydrogen as a fuel	▪ Can be produced from **diverse domestic resources** ▪ **Potential for near-zero greenhouse gas emissions**	▪ **Storing hydrogen is difficult** as it has a lower volumetric energy density ▪ **High production cost**
Biodiesel	▪ **Lean-burning, renewable substitute** for petroleum diesel. ▪ **Improves fuel lubricity** and raises the cetane number of the fuel. ▪ It is **safer than petroleum diesel** because it is less combustible.	▪ **Higher viscosity, lower energy content, higher nitrogen oxide (NO x) emissions,** lower engine speed and power.
Electricity	▪ **Improve fuel economy,** lower fuel costs, and reduce emissions.	▪ **Public charging stations are not as ubiquitous as gas stations.** ▪ The advanced batteries in electric vehicles are **designed for extended life but will wear out eventually.**
Natural Gas	▪ **Domestic availability,** established distribution network, relatively low cost, and emissions benefits.	▪ **Driving range of Natural gas vehicles (NGVs)** is generally less than that of comparable gasoline and diesel vehicles

Fuel Cell

- Electrochemical devices to convert chemical energy into electrical energy.
- Fuel cells **do not need to be periodically recharged** like batteries, but instead continue to produce electricity as long as it is supplied with a source of hydrogen and a source of oxygen (usually air).
- Source of **hydrogen is generally referred to as the fuel** and this gives the fuel cell its name.

- **Applications:** Transport, Portable (military applications, Auxiliary Power Units, laptops etc.), Stationary (power to industrial/commercial/residential buildings etc.),

Renewable Energy

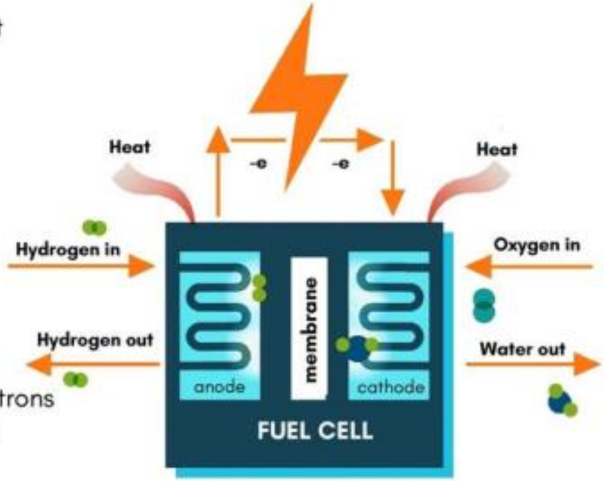
- **Benefits:** Lower operational costs due to higher efficiency and high energy density of hydrogen, Noise free operation, operating times are much longer.

- **Challenges:** Transportation, flammability and storage of hydrogen gas, High cost, Technological challenges like system size, management of air, heat and water etc.

Renewable Energy

How Fuel Cells Work

1. The hydrogen atoms enter at the **anode.**
2. The atoms are stripped of their electrons.
3. The positively charged protons pass through the membrane to the **cathode.**
4. The negatively charged electrons are forced through a circuit, generating electricity.



5. After passing through the circuit, the electrons combine with protons to generate the fuel cell's byproducts: **water** and **heat**.

Roshini

- It is India's **first Saline Water Lantern** which uses sea water as electrolyte between specially designed electrodes to power the LED lamps.
- Developed by: **National Institute of Ocean Technology (NIOT), Chennai.**
- Technology can also be used in hinterlands, as any saline/normal water mixed with common salt can be used to power the lantern.

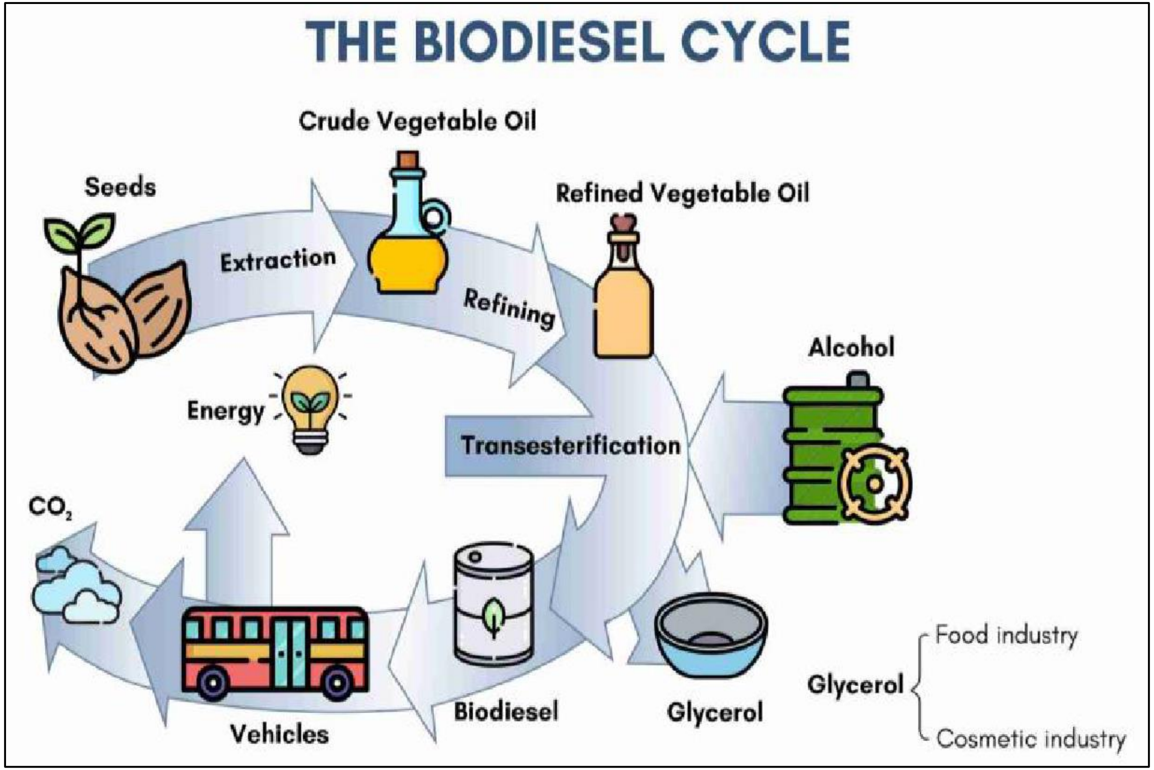
Biobutanol

- Biobutanol is **four-carbon alcohol** produced by the fermentation of biomass.
- The production of biobutanol can be carried out in ethanol production facilities.
- Its properties are **similar to that of gasoline.**
- Some gasoline-powered vehicles can even use biobutanol without being modified.
- However, it has a lower energy content, on average 10-20%, than that of gasoline, which is a major **disadvantage** of biobutanol.
- Biobutanol exhibits the potential to reduce carbon emissions by **85% when compared to gasoline,** thus making it a viable and suitable alternative to gasoline and gasoline-ethanol blended fuels.

Renewable Energy

- **Biodiesel**
- Biodiesel is made from renewable sources such as **vegetable/plant/animal oils for use in diesel engines.**
- Vegetable oils are chemically called **triglycerides** (fats).
- Biodiesel comprises esters of **long chain fatty acids** derived from these oils.
- To make biodiesel, fats in the vegetable oil (triglycerides) are reacted with alcohol — usually **methanol**.
- In this reaction, glycerine (in triglycerides) is replaced by methanol to produce **methyl ester (biodiesel).**
- Biofuel development in India centres around the cultivation of Jatropha plant seeds — rich in oil (40%).

Renewable Energy



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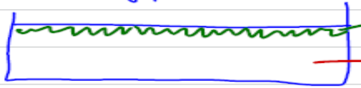
Biofuel from Microalgae: (3rd generation)

Algae → single celled
→ Sunlight → Algae (High lipid content)

Steps: Cultivation
- Raceway pond → algae ← Nutrient from H_2O ← Sun ← CO_2



Algae
No algae here, so lots of surface area needed

- Bioreactor (more expensive, but more efficient)

usually 1% by weight

↓

Harvesting (dewatering) 1% by weight ⟶ 50% by weight

Methods: floatation
Reverse osmosis

Lot of energy needed, expensive

Renewable Energy

3. Extraction → Mechanical
Chemical (pH)

fat floats out in water → separate from water → oil

oil → triglyceride
↳ HC backbone (ester)
+
3 glycerine

4. Transesterification of oil
↳ oil → breakdown c̄ catalyst (Methanol) → glycerine
→ HC Backbone (Biodiesel)

issues :
(1) Price
(2) Land availability (Raceway ponds)
(3) Fuel efficiency

Renewable Energy

- **Biodiesel blend**
- The biodiesel blend is a mix of biodiesel with fossil fuel, designated as BXX, where XX represents the volume percentage of biodiesel in the blend (B100 means pure biofuel).
- Currently, biodiesel is blended with **the fossil fuel in the proportion of 2%, 5%, etc,** although technically it can be used as a pure fuel with some minor modifications in existing engine systems.
- **Advantages of biodiesel**
- Biodiesel has intrinsic lubricating properties (diesel engines are long lasting compared to petrol engines because of this very property)

- The **Cetane Index (CI)**, a measure of the inflammability of fuel, is more than 56 to 58 for biodiesel compared 50/52 for fossil fuel. A higher CI value will mean better ignition and combustion.

About Octane Rating

- **Octane rating, also known as Octane number or Octane Value,** is defined as the percentage or volume fraction of isooctane in a mixture of isooctane and normal heptane fuel where knock is initiated at the same compression ratio as in the fuel.
- It measures the **fuel's ability to resist unwanted sounds due to auto-ignition with higher octane number.** E.g., adding ethanol to petrol helps in reducing knocking as ethanol octane rating is around 109.
- For diesel fuel, **Cetane number** is used to measure the ignition delay property of the fuel, with higher cetane number meaning reduced ignition delay to avoid knocking.

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- **Tetraethyl lead (TEL)**
    - Also called **Organic lead**, is used as a petrol additive in leaded petrol to improve engine performance.
    - It is a **colorless liquid** whose **antiknock** properties were first found in 1921.
    - It improves the **octane rating** of fuel as compared to unleaded petrol
    - As a result, it became a **popular additive** in petrol and jet fuels as knocking in engine causes loss of power with risks of damage to the engine.

- **Health Impacts of Leaded Petrol**
  - Tetraethyl lead is **toxic** in nature, and it is absorbed rapidly by the skin, the lungs, and the gastrointestinal tract.
  - It contaminates the air, dust, soil, water, and crops on release through exhaust fumes, evaporation losses and accidental spills.
  - Exposure to it can cause **Heart disease**, Cancer, stroke, and **lower IQ** (especially in children) by impacting brain development.
- Recently, leaded petrol has been eradicated from the world as **Algeria**- the last country to use this fuel, exhausted its supplies following the two decades long campaign by the UNEP-led global **Partnership for Clean Fuels and Vehicles (PCFV).**

Renewable Energy

- The biodiesel molecule contains about **11% oxygen**, facilitating improved combustion and less soot.
- The sulphur content in biodiesel is **as low as 0.001%.**
- Biodiesel requires less energy to produce than fossil fuels (for every unit of energy needed to produce biodiesel, 3.24 units of energy are gained)
- Biodiesel production does not hurt the production of edible oil ('food or fuel' dilemma doesn't exist).

- **Used Cooking Oil (UCO) based Biodiesel**
- **Context-** Recently, Indian Oil Corporation has kicked-off the supply of diesel that is blended with biodiesel made from used cooking oil.
- **About Used Cooking Oil (UCO)**
- UCOs are oils and fats **that have been used** for cooking or frying in the food processing industry, restaurants, fast foods and at consumer level, in households.
- UCO must contain **only fats, oils, or greases** that were previously used for cooking or frying operations.
- UCO is an important source of raw material to produce biodiesel.
- It is also used for making soap, cosmetics, cooking oil, and animal feed, etc.

Renewable Energy

- **<u>Benefits of biodiesel made from UCO</u>**
- **<u>Prevent people from serious ailments</u>** such as hypertension, obesity, atherosclerosis, Cancer, Alzheimer's disease, liver diseases by removing reused or burnt cooking medium from the food chain.
- UCO when discarded without any treatment **clogs drainage systems.**
- Contributing to the needs of a **Circular Economy.**
- **Raises chemical oxygen demand in water bodies.**
- **<u>Initiative taken to make biodiesel from UCO</u>**
- In 2019, Ministries of Petroleum and Natural Gas & Steel, along with Health & Family Welfare, Science & Technology and Earth Sciences, had initiated Expressions of Interest for "Procurement of Bio-diesel produced from UCO"

Renewable Energy

- It aims to create an ecosystem for collection and conversion of UCO into Biodiesel and developing entrepreneurship opportunities.
- Under this initiative**, OMCs offer periodically incremental price guarantees for five years** and extend off-take guarantees for ten years to prospective entrepreneurs.
- **National Policy on Biofuels, 2018,** encourages setting up of supply chain mechanisms for biodiesel production from non-edible oilseeds, Used Cooking Oil, short gestation crops.
- **Food Safety and Standards Authority of India (FSSAI)** in association with the **Biodiesel Association of India (BDAI)** launched ‘**Repurpose Used Cooking Oil (RUCO) project**’ in 2019.

Renewable Energy

- Project is aimed at purchasing used oils from hoteliers, caterers, snack makers and traders at a reasonable price and converting it into biodiesel at a plant.
- RUCO sticker and a mobile phone application was launched for collection of used cooking oil (UCO) to ensure that it does not come back to ecosystem.
- Another scheme launched for UCO based biodiesel : **Randhan se indhan (cooking to fuelling)**

- **<u>Bio-Jet Fuel Technology</u>**
- **Context- CSIR-IIP Dehradun's** home-grown technology to produce bio-jet fuel, formally approved for **use on military aircraft of the Indian Air Force.**
- Bio-jet fuel can be produced from used cooking oil, tree-borne oils, short gestation oilseed crops, and waste extracts from edible oil processing units.
- Earlier, AN 32 (transportation plane) and commercial plane (Spice jet) was flown in 2018 using the bio-jet fuel.

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Bio Gas

- **Primarily methane and carbon dioxide**. Produced through a process of anaerobic decomposition from biomass.
- **anaerobic digestion uses fermentation to breakdown organic matter**. after purification and compression it is called compressed biogas.
- CBG has **pure methane content** over 95%.
- It is **similar to CNG in its composition** and energy potential.
- Biogas production is **carbon neutral**.

Compressed BioGas (CBG)

- **Context:** Union Minister of Petroleum and Natural Gas inaugurated Asia's largest CBG plant in Sangrur, Punjab.

- CBG is the compressed and purified bio-gas produced from waste/bio-mass sources like agricultural residue, cattle dung etc.

- **Benefits:**
  - **Reduce the problem of Stubble Burning** and **related carbon dioxide emissions.**
  - Produce fermented organic manure for organic farming.
  - Generate employment and additional income for farmers.

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Sustainable Alternative Towards Affordable Transportation (SATAT)

- It is an initiative launched by the **Ministry of Petroleum & Natural Gas** and Skill Development & Entrepreneurship in October 2018.
- The initiative aims to set up **Compressed Bio-Gas (CBG)** production plants and make available CBG in the market for use in automotive fuels.
- The **objective** of SATAT initiative is to boost the availability of more **affordable transport fuels**, better use of **agricultural residue**, cattle dung and municipal solid waste as well as to provide an additional revenue source to farmers.
- SATAT was launched in association with the **Public Sector Undertakings** and Oil Marketing companies
- The Compressed Bio-Gas plants are proposed to be set up mainly through **independent entrepreneurs**.
- It is planned to roll out **5,000 Compressed Bio-Gas plants** across India in a phased manner in the next 4 years.

Biomass co-firing

- **Context:** Government had launched the **SAMARTH Mission** i.e., National Mission on Use of Biomass in coal based thermal Power Plants.

- **Co-firing biomass** consists of burning biomass in coal-fired power plants along with coal. Three different concepts for co-firing i.e, Direct co-firing, Indirect co-firing, Parallel co-firing.

- **Advantages: lower** capital costs, higher efficiency, increased economies of scale, **Lower electricity costs** due to the larger size and superior performance of conventional coal power plants

- **Direct co-firing –** The biomass and the coal are burned in the same furnace. The mills for the grinding of the fuel and the burners may be the same as or be separate from the coal feeding.
- **Indirect co-firing –** In this concept, the solid biomass is converted to a clean fuel gas, using a biomass gasifier. The gas can be burnt in the same furnace as the coal. For this reason, it is also possible to use biomass, which, for example is difficult to grind. This principle is less used compared to direct co-firing.
- **Parallel co-firing –** It is also possible to install a completely separate biomass boiler in addition to the conventional boiler. In this case, the biomass boiler provides energy to the feed water or generates steam at low temperature, while the conventional boiler tops up the superheat.

Renewable Energy

- **National Bioenergy Programme (NBP)**
- Context
- Recently, Ministry of New and Renewable Energy (MNRE) has notified that National Bioenergy Programme (NBP) will continue from FY 2021-22 to 2025-26 with a budget outlay of ₹858 crore for the first phase.
- NBP will comprise the three sub-schemes

1. Waste to Energy Programme
2. Biomass Programme
3. Biogas Programme.

Renewable Energy

- **Terms**
- **Biomass** is renewable organic material that comes from plants and animals. Biomass is the raw material to produce biogas. Major component of biomass is **carbon**. [Recall carbon cycle]
- **Bioenergy** refers to electricity and gas that is generated from organic matter, known as biomass.
- **Biogas** is a renewable fuel produced by the breakdown of organic matter such as food scraps and animal waste by microorganisms in absence of oxygen.
- Biogas is composed mostly of **methane (CH_4),** and **carbon dioxide (CO_2).**

Renewable Energy

- **Syngas:** Mixture of CO and H_2 (obtained mainly from natural gas)
- **Water gas:** Consists of CO and H_2 (obtained mainly from coke)
- **Bio gas:** Methane (40%–75%) + Carbon dioxide (15%–60%)
- **Liquefied Petroleum Gas:** Mixture of propane (C_3H_8) and butane (C_4H_{10})
- **Compressed natural gas (CNG)**: Mainly Methane (CH_4)

Renewable Energy

- **Sub-scheme 1: Waste to Energy Programme (Programme on Energy from Urban, Industrial and Agricultural Wastes /Residues)**
- Objective: To support the setting up of **Waste to Energy** projects for generation of Biogas/ Bio-CNG/ Power plants (excluding municipal solid waste to power project)/ producer or syngas from urban, industrial and agricultural wastes/residues.
- **Implementing agency:** Indian Renewable Energy Development Agency (IREDA).
- IREDA, established in 1987, is a **Mini Ratna** (Category-I) Enterprise under administrative control of **MNRE**.

Renewable Energy

- **Scope**
- Provides Central Financial Assistance (CFA) to project developers and service charges to implementing/inspection agencies in respect of successful commissioning of waste to energy plants.
- **Advantages**
- Safe, technologically advanced means of waste disposal that reduces greenhouse gases, generates clean energy and recycles metal.

- **<u>Sub-scheme 2: Biomass Programme</u>**
- Also called Scheme to Support Manufacturing of Briquettes & Pellets and Promotion of Biomass (nonbagasse) based cogeneration in Industries
- Objective
- To support setting up of Biomass Briquette/Pellet manufacturing plants and to support Biomass (non-bagasse) based cogeneration projects in Industries.
- [A briquette is a compressed block of coal dust or other combustible biomass material]
- Implementing agency: IREDA.

Renewable Energy

Renewable Energy

- **<u>Scope</u>**
- Provides Central Financial Assistance (CFA) to project developers and service charges to implementing agency and inspection agencies in respect of setting up of Briquette / Pellet manufacturing plants and Biomass (non bagasse) cogeneration projects in industries.
- **<u>Advantages</u>**
- Reduce stubble burning. [More discussion to be followed while discussing stubble burning]
- Provide additional source of income to farmers and better environmental practices and reduced pollution.

- **<u>Sub-scheme 3: Biogas Programme</u>**
- Objective
- Setting up of biogas plants for small power needs of users.
- Implementation
- By designated Programme Implementing Agency (PIA) of State/ Union Territory/Biogas Development and Training Centres (BDTC).
- IREDA, NABARD/ RBI approved Financial Institutions may also implement it in consultation with PIAs.

Renewable Energy

- **Scope**
- Covers setting up of small and medium biogas plants ranging from 1M3 (cubic meter) to 2500 M3 biogas generation per day for individual user, farmers, poultry etc.
- **Advantages**
- GHG reduction, improved sanitation, Employment Creation, and digested slurry from biogas plants can supplement / reduce use of chemical fertilizers.

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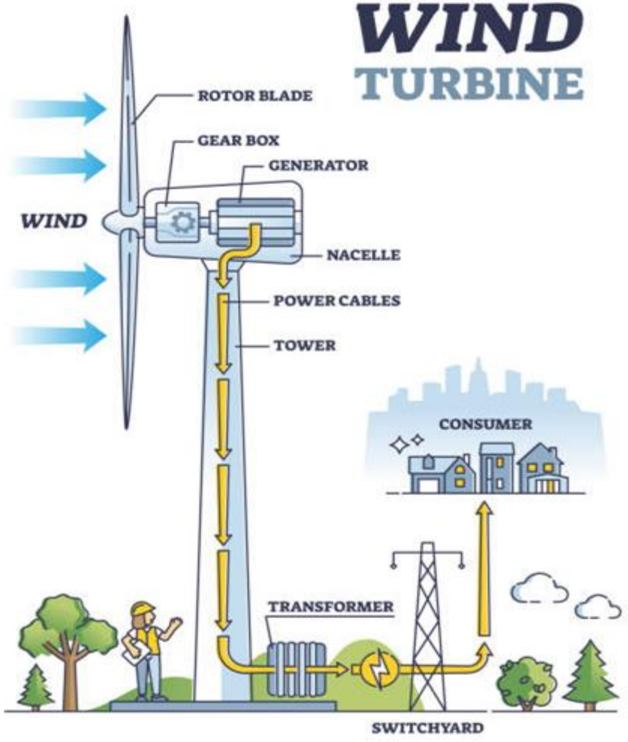
Wind Energy

- **Principle:** Kinetic energy of wind to mechanical energy of rotor blades which can move turbine-generator to produce electrical energy.
- It generates **AC power**
- **Gear Box:** Converts low Revolutions per minute (RPM) to high RPM.



Suitable locations for Wind Energy Plants

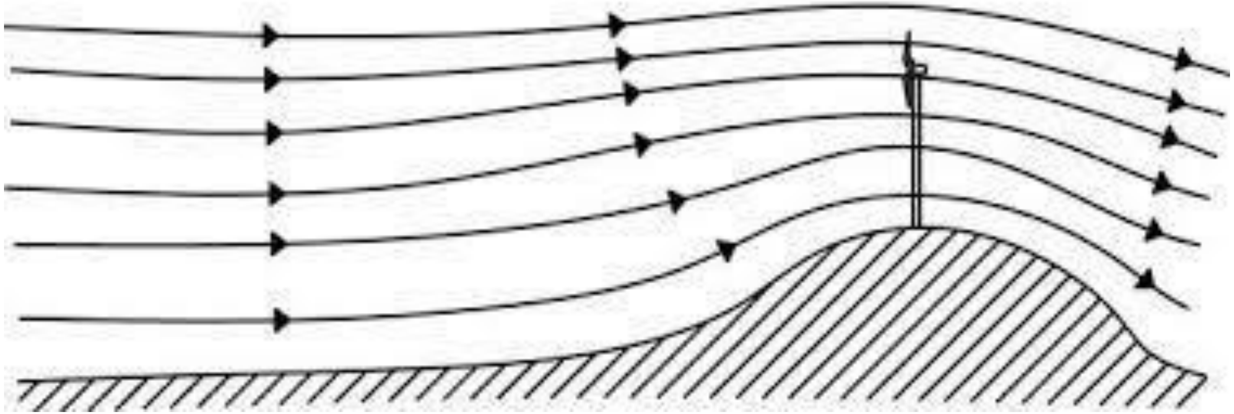
1. <u>Coastal</u>

- **Onshore:** Most of the India's wind power is on shore.
- **Offshore:** They generate electricity from wind blowing across the sea. They are more efficient than onshore wind farms, due to the **higher speed of winds, greater consistency and lack of physical interference.**
  - **Pilot Projects:** Kanyakumari, Rameshwaram, etc.

Suitable locations for Wind Energy Plants

2. Hilly Region

- Favorable sites include the **tops of smooth, rounded hills and mountain gaps that funnel and intensify wind.**



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Suitable locations for Wind Energy Plants

3. Islands

- Favorable sites include the **tops of smooth, rounded hills and mountain gaps that funnel and intensify wind.**
- **Andaman Nicobar Islands (longer islands) > Lakshadweep (atolls)**

4. Deserts:

- **Example: Saudi Arabia** has a potential of **200 GW** due to its Desert landscape.

Wind Aero generator

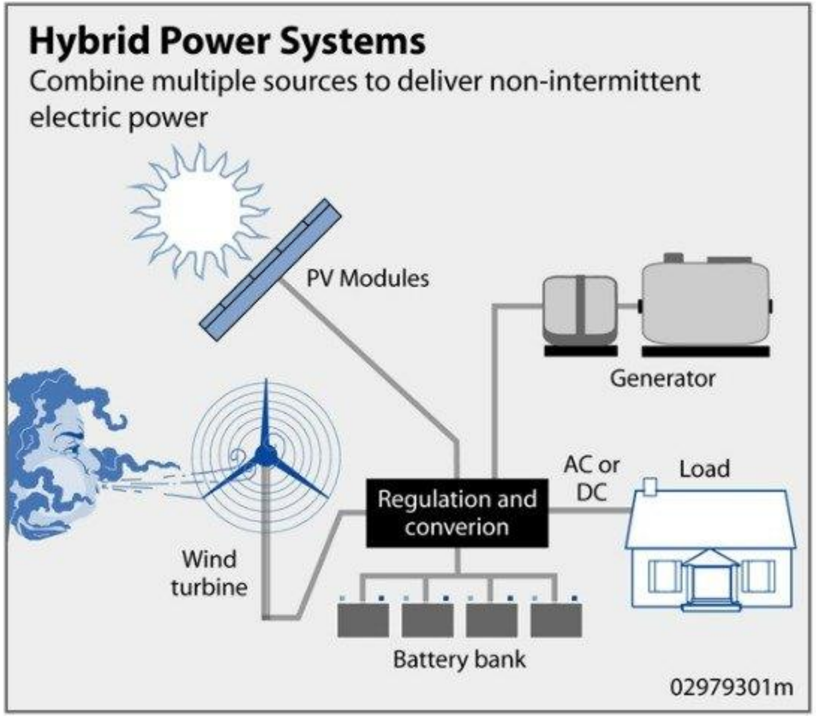
- An aerogenerator is a **small wind electric generator** having a capacity of up to **30 kW.**
- Aerogenerators are installed either in stand-alone mode (off grid) or along with solar photovoltaic (SPV) systems to form a wind-solar hybrid system for **decentralized power generation.**
- An aerogenerator is suitable for power generation in unelectrified areas having adequate wind speeds.

Wind-Solar Hybrid System

- A wind solar hybrid project is a renewable energy system that combines both wind turbines and solar panels in the same location to generate electricity. This approach is a solution to the **intermittency challenges** of both solar and wind energy generation providing a more stable and reliable energy output.

- The **MoNRE** has issued the **National Wind-Solar Hybrid Policy** in **2018** with an aim to boost renewable power generation by promoting new projects as well as hybridisation of existing ones.

Renewable Energy



Renewable Energy

- **<u>Wind Energy</u>**
- India currently has an installed capacity of 41 GW (Feb 2023), the fourth highest in the world. **(China> USA> Germany > India)**
- However, this is just **29%** of its target, which is to install **140 GW by 2030.**
- Tamil Nadu accounts for **25%** of this installed capacity followed by Gujarat **at 24%.**
- **Important wind farms**
  - Muppandal Wind Farm, Kanyakumari district- 1.5 GW
  - Jaisalmer Wind Park, Rajasthan- 1.06 GW

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Muppandal Wind Farm, India

- The area of Muppandal features vast barren land, which is not fit for cultivation but **boasts of high pressure winds (due to funnelling effect)** making it an ideal location for the development of wind farms.
- Muppandal receives strong winds for nine months a year from the west, due to the presence of the **Western Ghats mountain range.**

National Offshore Wind Energy Policy

- It has been notified in **2015** to provide a legal framework for development of the offshore wind sector in India.
- Policy authorises, the **Ministry of New and Renewable Energy** to work in close coordination with other government entities for development of offshore wind energy within the **entire exclusive economic zone (EEZ)** of the country.
- **National Institute of Wind Energy (NIWE)** has been designated as the nodal agency to carry out or facilitate necessary resource assessment activity for offshore wind development.

Global Offshore Wind Alliance

- Nine new countries including Belgium, Colombia, Germany, US, UK etc. join GOWA.
- Initiated by: **International Renewable Energy Agency (IRENA), Denmark and Global Wind Energy Council (GWEC).**
    - GWEC is international trade association for the wind power industry (India is also a member).
- Brings together governments and other stakeholders to accelerate deployment of offshore wind power.
- **Objective:** To contribute to achieving a total offshore wind capacity of a minimum of 380 GW by 2030 and an installed capacity increase of at least 70 GW per year from 2030.

Small Hydro Power

- In India, hydro power plants with **capacity of 25 MW or below** are classified as small hydro.
- Hydro Power was looked after by **Ministry of Power** prior to **1989.**
- In **1989,** plant capacity up to 3MW and below was transferred to the Ministry of New and Renewable Energy (MNRE). And, Subsequently plant capacity up to **25MW** and below was entrusted with the MNRE in November **1999.**
- **Estimated potential:** 21133 MW from 7133 sites

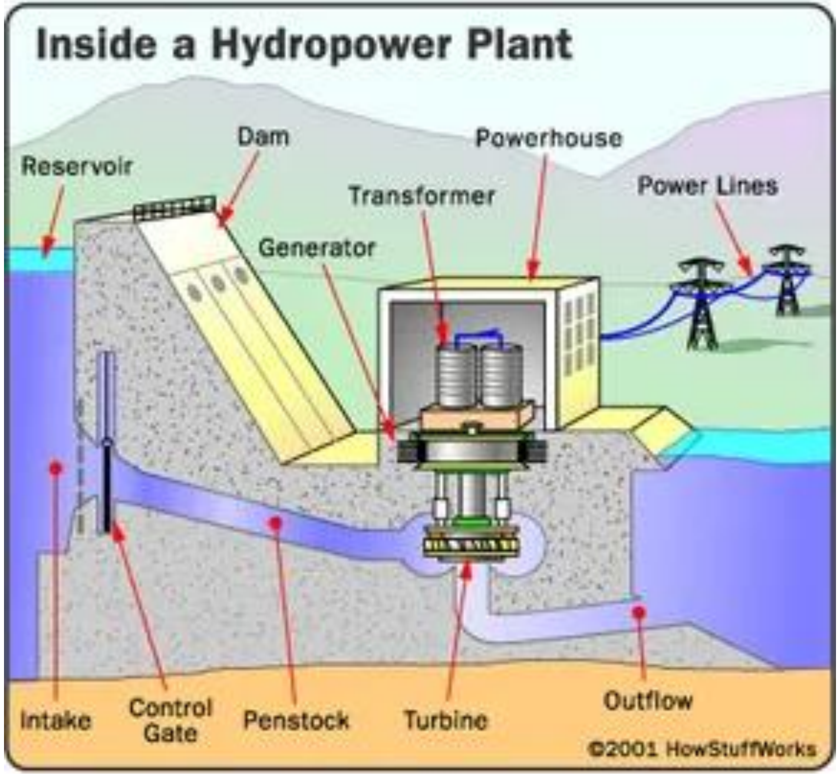
Renewable Energy

- The hydropower facilities are **powered by the kinetic energy** of flowing water as it moves downstream.

- Hydropower **utilizes turbines and generators** to convert that **kinetic energy into electricity,** which is then fed into the electrical grid to power homes, businesses, and industries.

- At the plant level, water flows through **a pipe—also known as a penstock**—and then spins the blades in a turbine, which, in turn, spins a generator that ultimately produces electricity.

Renewable Energy



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Advantages:

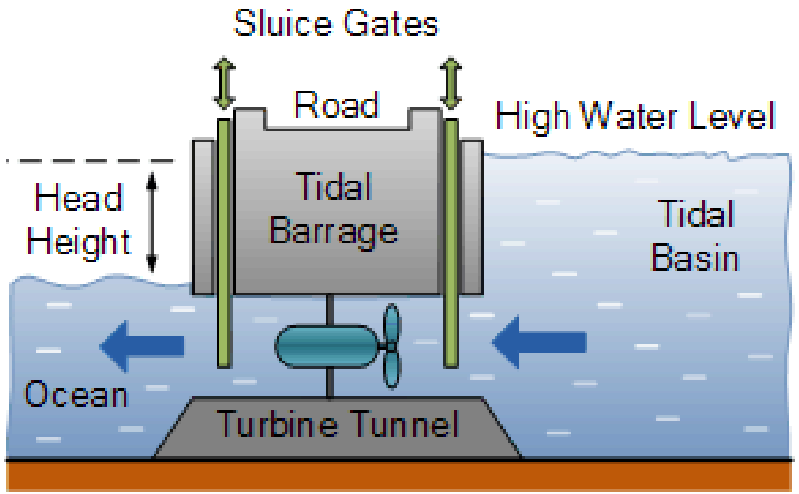
- Can be used in difficult terrain particularly hilly areas for power supply
- Lese ecological impact

Disadvantages:

- Less power generation of **<25 MW**
- Only **Local** application (not to grid)
- **Seasonal** in nature (especially & non perennial rivers).
- If water NOT returned to river and used for irrigation, may lead to **water loss in lower riparian areas**

Tidal Energy

- **Mechanism**:
- High tide- Water flows from sea to tidal basin. **(moves turbine to generate electricity)**
- Low tide: Water goes back.



- **Requirements:**
  - Continuous high and low tide at **regular intervals.**
  - Ideally, There should be **two High and low tides** each in a day.

Renewable Energy

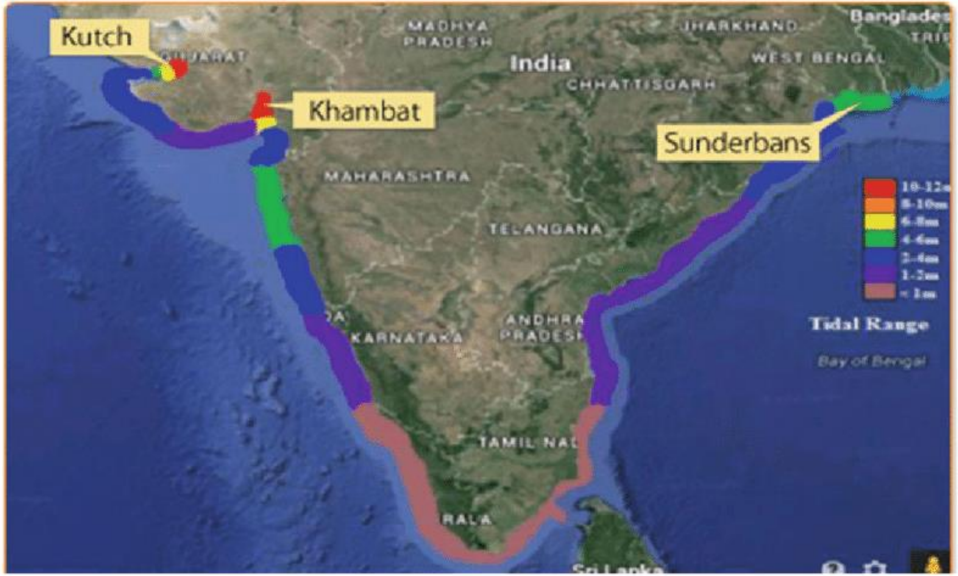
- **Tidal Range: Minimum height difference between HT and Lt: 5 meters.**

- **Locations**: Creeks, Gulfs, Estuary, Delta.

- **Locations in India:** Gulf of Kutch, Gulf of Khambat, Sundarbans Delta (**Durgaduani creek project:** stopped)

- **Worldwide:**
  - Only 20 locations in world where tidal power is generated.
  - **Oldest:** Rance power project (240 MW)
  - **Largest:** South Korea (Sihwa lake- 254 MW)

Renewable Energy



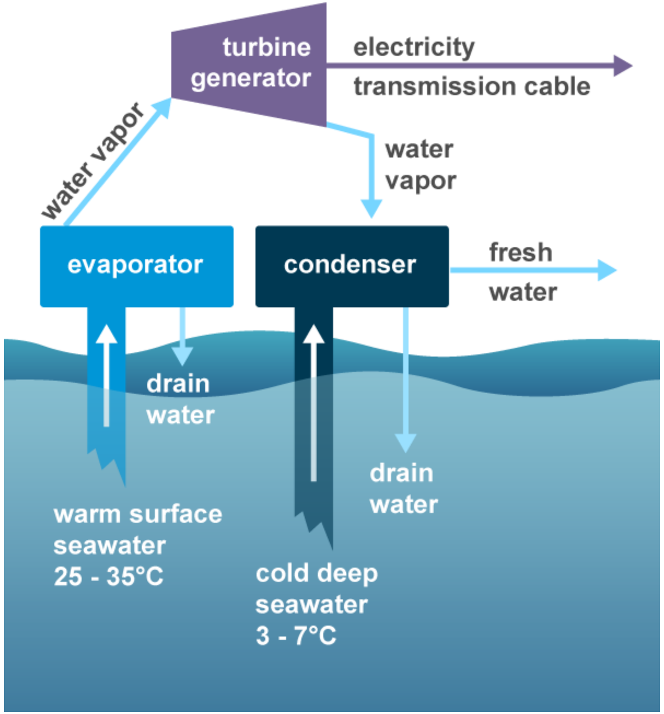
Gulf of Cambay/Khambat	Gulf of Kutch	Sunderbans Delta
• Maximum Tidal Range: 11m • Average Tidal Rang: 6.77 • Potential: 7000MW	• Maximum Tidal Range: 8m • Average Tidal Rang: 5.23 • Potential: 1200MW	• Maximum Tidal Range: 5m • Average Tidal Rang: 2.97 • Potential: 100MW

Ocean Thermal Energy Conversion (OTEC)

- **Mechanism:**
  - A process for producing energy by harnessing the **temperature differences (thermal gradients)** between ocean surface waters and deep ocean waters. o In tropical regions, surface water, heated by Sun's energy, can be much warmer than deep water. This temperature difference can be used to **produce electricity and to desalinate ocean water.**
  - OTEC systems use a temperature difference of **at least 20° Celsius** to power a turbine to produce electricity.

Renewable Energy

- Hot surface water is used to convert volatile liquid such as; Ammonia propane into gas in a heat exchanger.
- This gas turns the turbine and is condensed back to liquid ē cold deep water.
- An OTEC plant can generate **100 MW power.**



Two types of OTEC technologies:

- **Closed cycle method:** A working fluid, such as **ammonia, is pumped through a heat exchanger and vaporized.** This vaporized steam runs a turbine. The **cold water found at the depths of the ocean condenses the vapor back to a fluid** where it returns to the heat exchanger.

- **Open cycle system:** The **warm surface water is pressurized in a vacuum chamber and converted to steam to run the turbine. The** steam is then condensed using cold ocean water from lower depths.

- **India's potential:** OTEC has a theoretical potential of 180,000 MW in India subject to suitable technological evolution.

Advantages:

1) Carbon free
2) High capacity
3) Potential to generate potable water, hydrogen, and ammonia; Cold and nutrient rich water from the
4) OTEC process can benefit commercial products such as air conditioning, aquaculture, etc.

Disadvantages:

1) High investment
2) Corrosive Water
3) Affects marine life **(entanglement in cables, entrapment, noise generation, accidental release of biocides)**

Renewable Energy

In News

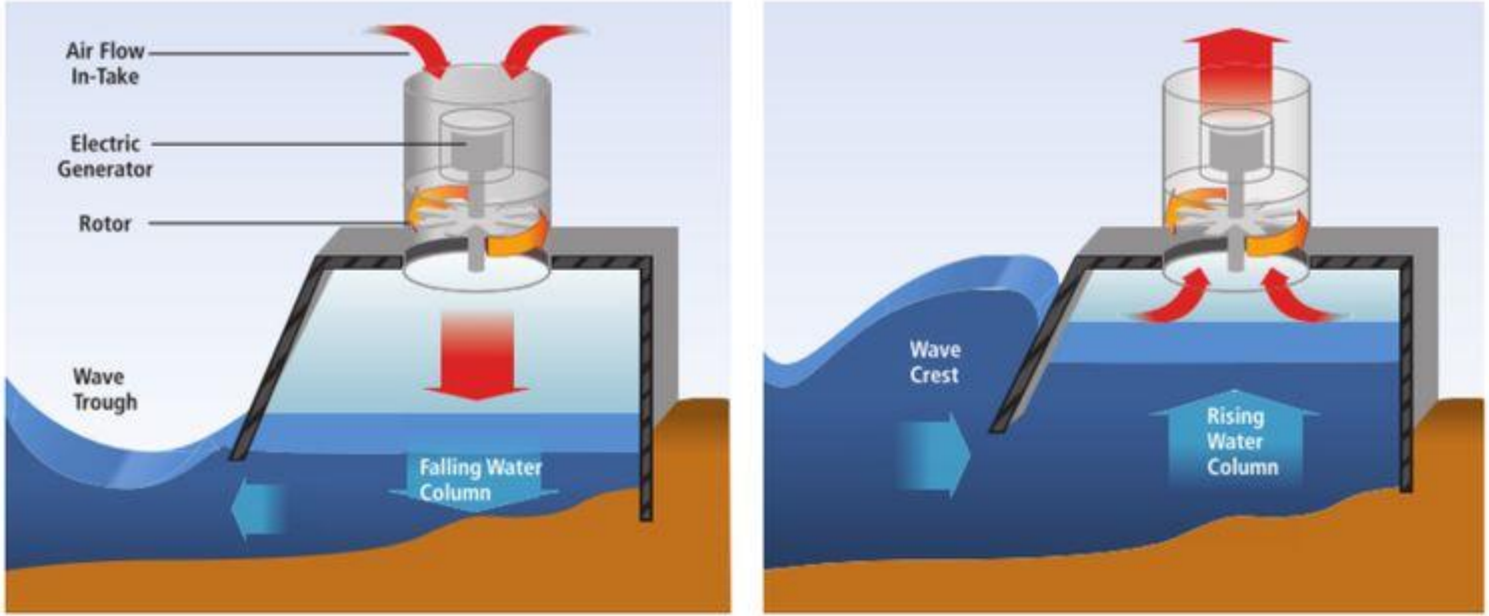
- The National Institute of Ocean Technology **(NIOT)**, an autonomous Institute under Ministry of Earth Sciences **(MoES)** is establishing India’s first **Ocean Thermal Energy Conversion (OTEC) plant.**
- **OTEC plant will have a capacity of 65kW in Kavaratti, Lakshadweep with indigenous technology.**
    - **It will power the Low Temperature Thermal Desalination (LTTD) based** desalination plant for conversion of Sea water into Potable water.
- Earlier in 1980, India **had planned to set up an OTEC plant, off the Tamil Nadu coast, but it was abandoned.**

Sindhuja-I (Ocean Wave Energy Converter)

- It is an Ocean Wave Energy Converter developed by **IIT Madras.** It consists of a floating buoy, a spar, and an electrical module.
- Electric module generates electricity by using the relative motion between floating buoy (moving up and down with waves) and fixed spar passing through buoy centre.
- Seawater stores tidal, wave and ocean thermal energy, make the harnessing of **40 GW wave energy** possible in India.

- **Wave energy converters (WECs)** are devices that convert the kinetic and potential energy associated with a moving ocean wave into useful mechanical or electrical energy.

Renewable Energy



Coal-bed Methane (CBM)

- CBM is an **unconventional form of natural gas** found in coal deposits or coal seams, formed during the process of coalification, transformation of plant material into coal. When coal and methane conversion process occur such that **coal is saturated with water and methane is trapped within coal,** the result is CBM.

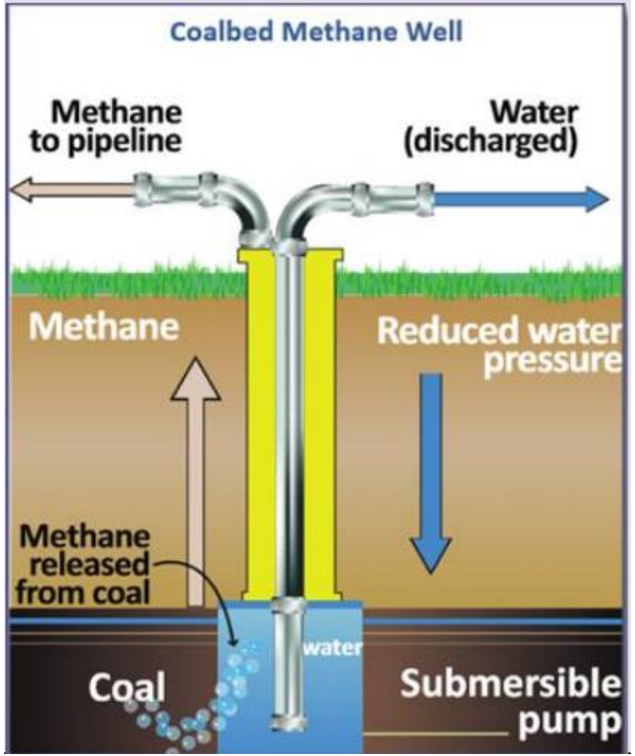
- **Uses:** Power generation, as CNG auto fuel, feedstock for fertilisers, industrial uses such as in cement production, rolling mills, steel plants, and for methanol production.

- Ministry of Petroleum & Natural Gas formulated CBM Policy under Oilfields (Regulation & Development) Act 1948 and Petroleum & Natural Gas Rules 1959 to harness the potential of CBM.

Process of Extraction

- A **steel-cased hole** is drilled into the coal seam and the underground (produced) water is pumped out through tubing.
- Removal of produced water **helps reduce the hydrostatic pressure within the coal bed,** causing the gas to be desorbed from its surface.
- As production occurs, the **change in pressure alters the porosity and permeability** of the coal bed.
- The recovered gas is sent into a natural gas pipeline or air **compressor system.**

Sustainable Development



Geothermal Energy

- ONGC has reportedly started drilling its first well at Puga (Ladakh) to generate electricity using geothermal power.

- Geothermal power plants use **hydrothermal resources that have both water (hydro) and heat (thermal).** Hot water and steam from deep underground can be piped up through underground wells and used to generate electricity in a power plant.

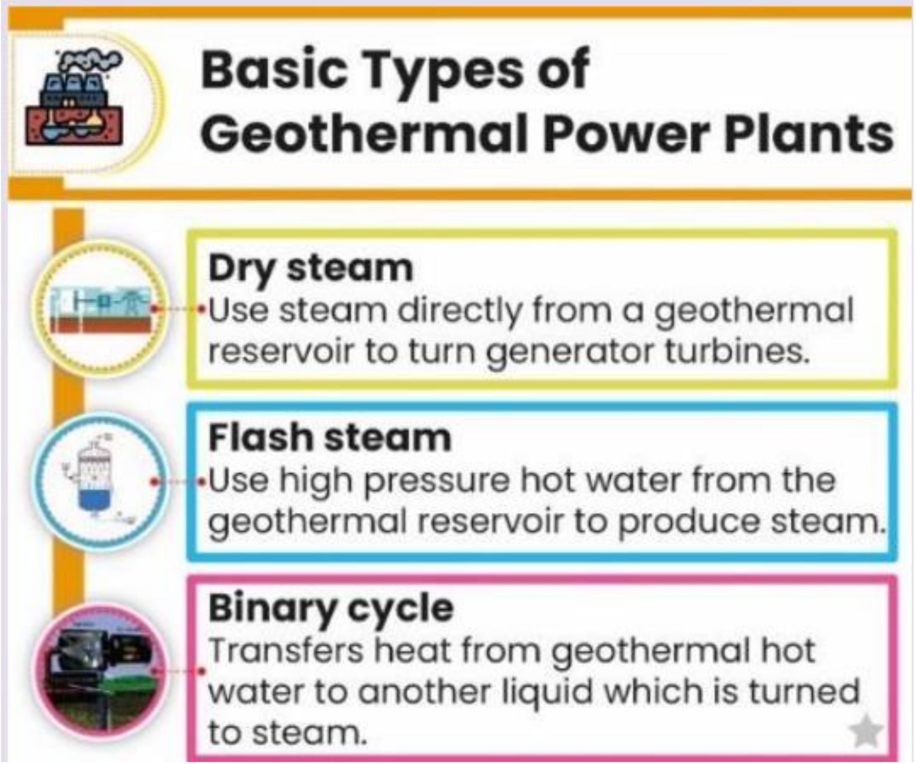
- **Benefits of Geothermal:** carbon-free, renewable, provides a continuous, uninterrupted supply etc.

- **Disadvantages:** Location-specific energy source, associated with other emissions like sulphur dioxide and hydrogen sulphide, causes mini tremors in area of operation, high initial cost etc.

Sustainable Development

- **Potential:** As per preliminary investigations undertaken by Geological Survey of India, there are around **300 geothermal hot springs** in India.

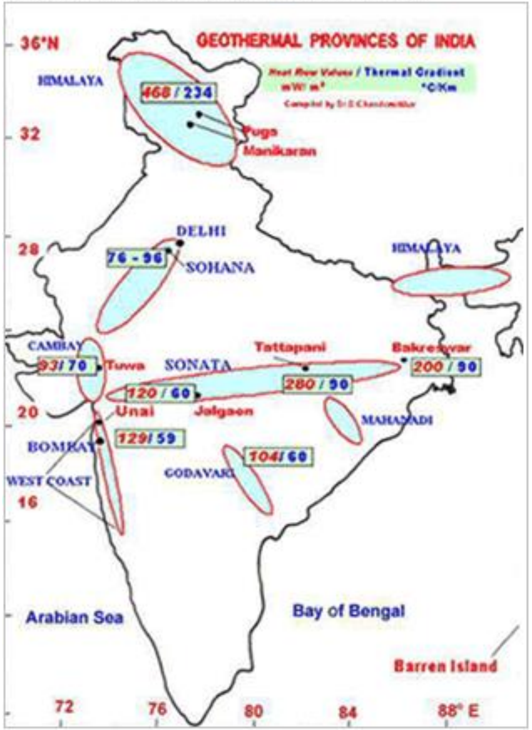
- **Geothermal sites (in decreasing order):**
  Tattapani in Chhattisgarh, **Puga** in J&K, **Cambay** in Gujarat, **Manikaran** in Himachal Pradesh, **Surajkund** in Jharkhand, **Chumathang** in Ladakh.



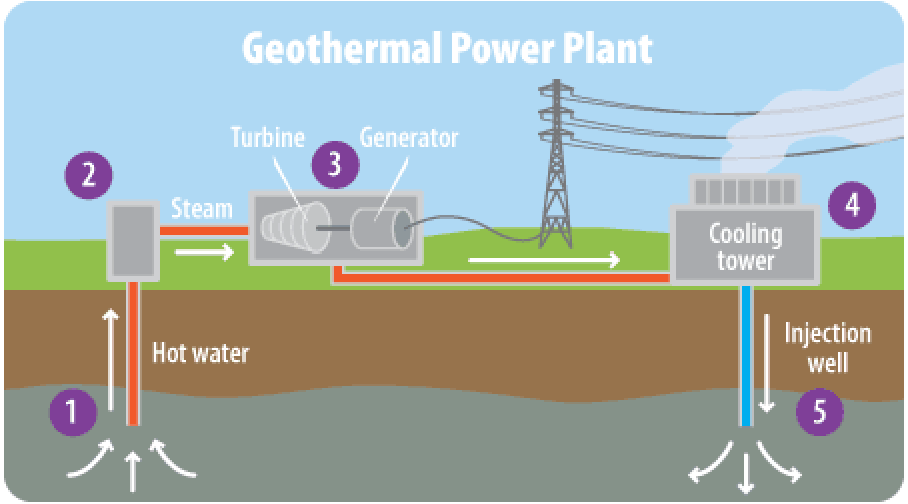
- **Geothermal Provinces of India**

1. Himalayan belt
2. Naga-Lushai Province
3. Andaman Nicobar Islands
4. West Coast Province
5. Cambay Graben Province
6. Aravalli Province
7. Son-Narmada -Tapi province
8. Godavari Valley Province
9. Mahanadi valley Province
10. South Indian Cratonic Province

Major geothermal energy resources in India

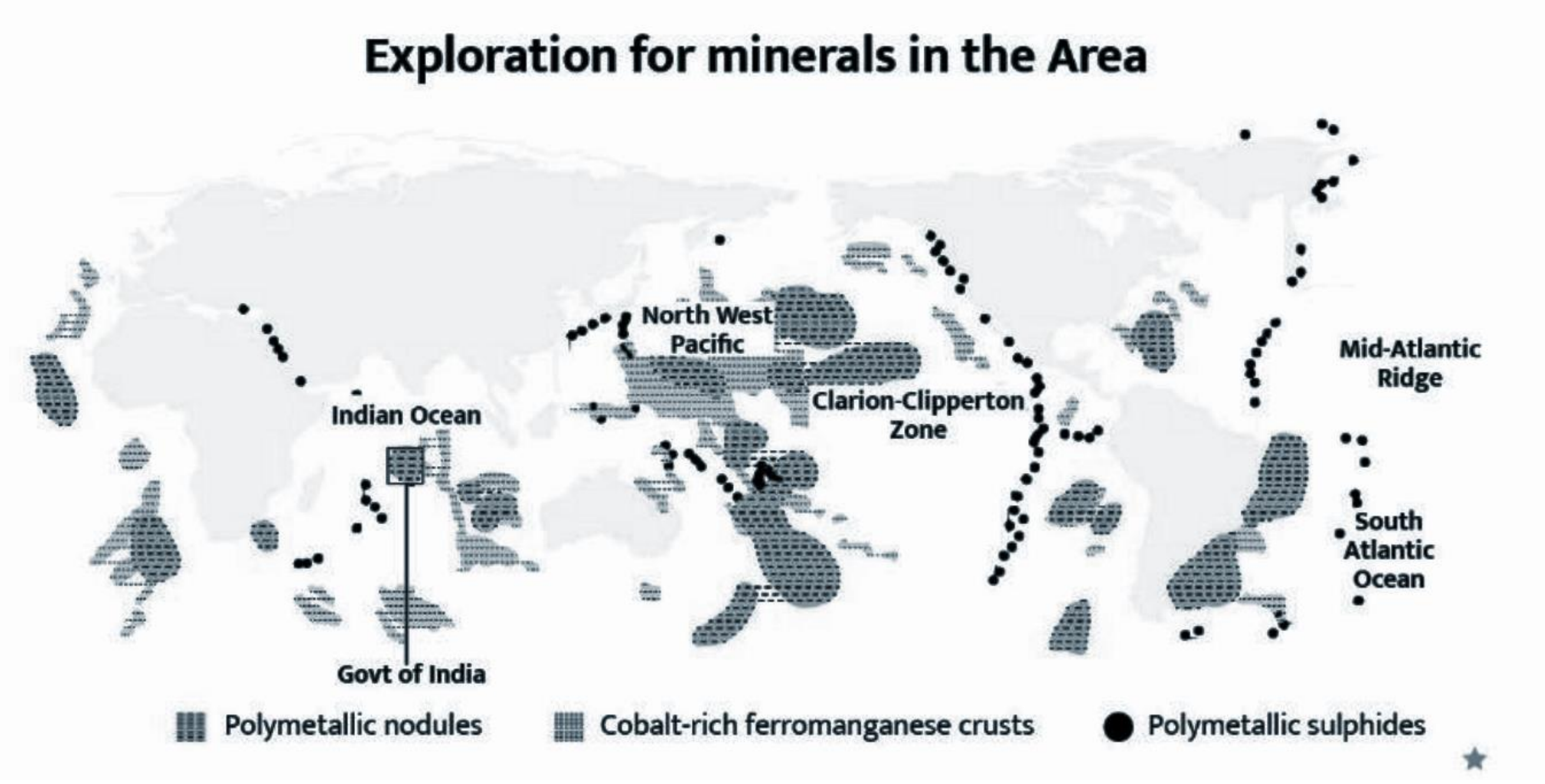


Geothermal well: used in power generation. Water circulated through a pipe in earth's crust and converted to steam. This Steam is used to run turbine for electrical power



Deep-Sea Mining

- Deep-Sea Mining involves extracting ores rich in cobalt, manganese, zinc and other rare metals from the sea floor of the deep ocean.
- India has a Deep-Sea mission for mining Central Indian Ocean Basin, access for which was granted by ISA.
- 54% of the world's ocean seafloor is beyond national jurisdiction and their mineral resources are designated as the common heritage of all mankind.
- Remember **UNCLOS : continental shelf claims happen to UNCLCS.**
- **Remember : High Seas Treaty**



Exploration for minerals in the Area
North West Pacific
Clarion-Clipperton Zone
Indian Ocean
Govt of India
Mid-Atlantic Ridge
South Atlantic Ocean
Polymetallic nodules
Cobalt-rich ferromanganese crusts
Polymetallic sulphides

Sustainable Development

- Deep-sea contains a number of critical minerals needed for batteries, smartphones, laptops and renewable energy capacity; found in three main resource types:
    - **Polymetallic Nodules,** i.e. the precipitated iron oxyhydroxides and manganese oxides on abyssal plain sediments at about 3,500–6,000 m water depths.
    - **Seafloor Massive Sulphides or Polymetallic Sulphides** formed along tectonic plate boundaries and volcanic provinces in water depths from <500 to 5000m.
    - **Cobalt-rich Ferromanganese Crusts** on rock outcrops on seamounts and ridges at water depths of 400–7,000 m.

International Seabed Authority (ISA)

HQ
Kingston, Jamaica

Genesis: An **autonomous international organisation** which came into existence in **1994** when **United Nations Convention on the Law of the Sea (UNCLOS), 1982** came into force.

Objective: To ensure the **effective protection of the marine environment** from harmful effects that may arise from deep-seabed related activities.

Membership: 168 members, including 167 member States and the European Union

Member

Other key information:
UNCLOS also established 2 other bodies namely, **International Tribunal for the Law of the Sea (ITLOS) and Commission on the Limits of the Continental Self (CLCS).**

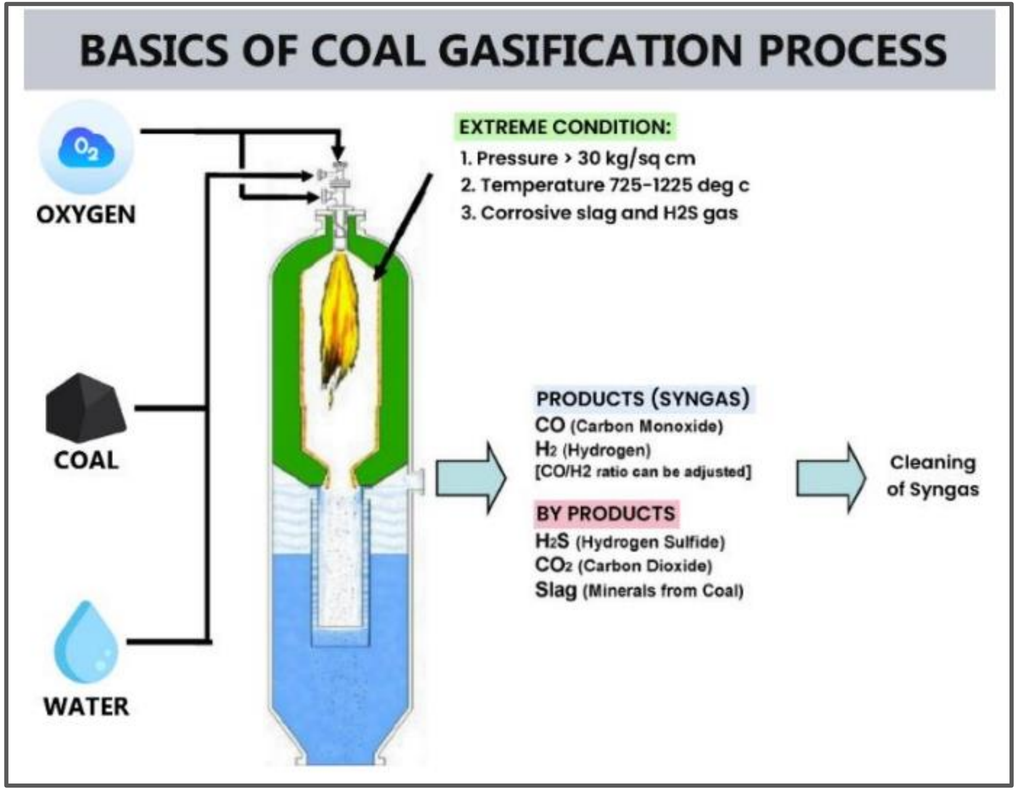
Coal Gasification

- **Context:** Union government has allowed a concession of 50% in revenue share for coal gasification.

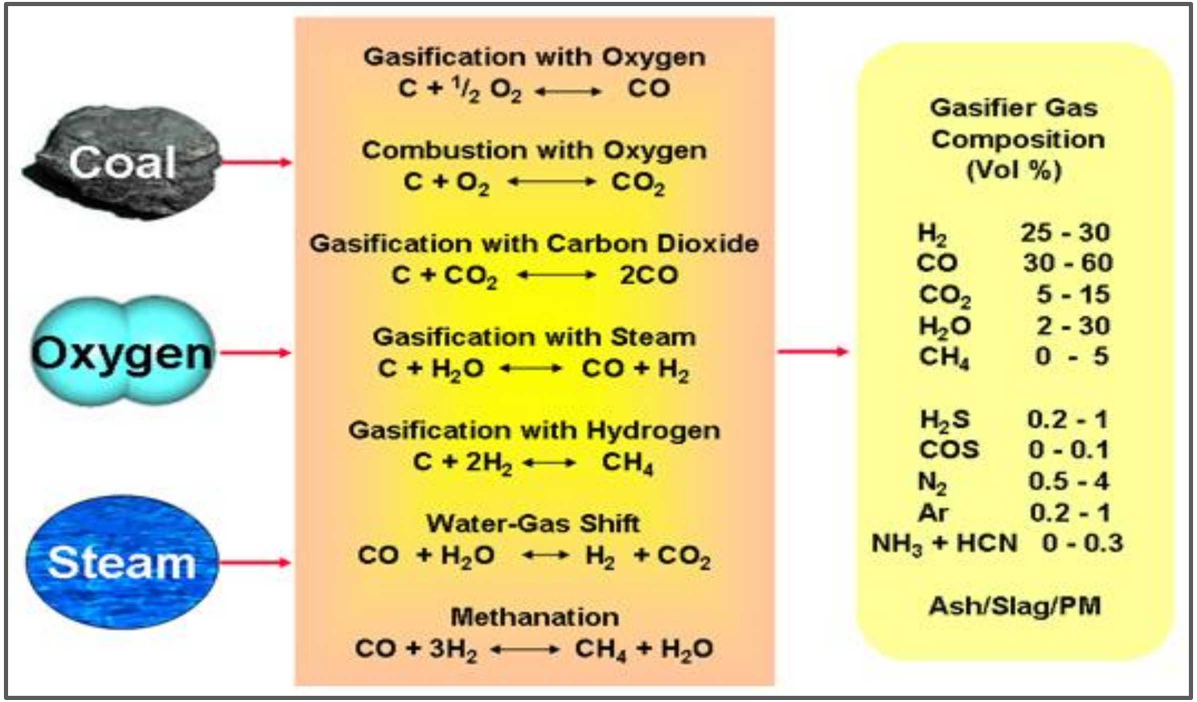
- Coal gasification (CG) is the process **converting coal** into **synthesis gas** (syngas), which is a mixture of hydrogen (H_2), carbon monoxide (CO) and carbon dioxide (CO_2).

Process of Coal Gasification

- Coal is fed into a vessel called **gasifier.** In gasifier, Coal is exposed to **heat and pressure.**
- **Controlled amount of oxygen** is released into gasifier allowing the coal to burn **(Partial oxidation)**
- Heat and pressure break up the structure of partially oxidized coal causing a chemical reaction to take place creating **syn gas.**
- **Ash and other byproducts** are trapped in the bottom of the gasifier.
- In the gasification process, **sulphur** present in the coal is converted to **hydrogen sulphide (H_2S)** and minor amounts of carbonyl sulphide (COS).
  - These sulphur compounds can be easily and economically removed using **acid gas removal systems.**


BASICS OF COAL GASIFICATION PROCESS
OXYGEN
COAL
WATER
EXTREME CONDITION:
1. Pressure > 30 kg/sq cm
2. Temperature 725-1225 deg c
3. Corrosive slag and H2S gas
PRODUCTS (SYNGAS)
CO (Carbon Monoxide)
H2 (Hydrogen)
[CO/H2 ratio can be adjusted]
BY PRODUCTS
H2S (Hydrogen Sulfide)
CO2 (Carbon Dioxide)
Slag (Minerals from Coal)
Cleaning
of Syngas

Sustainable Development



Coal Gasification

- **In-situ gasification process:** Underground coal gasification (UCG) is an in-situ gasification process. It is an industrial process which converts coal into product gas. It carried out in non-mined coal seams using injection of oxidants and steam. The product gas is brought to the surface through production wells drilled from the surface

- **Ex-situ gasification process:** In **ex-situ process**, reactor is developed for simulating the gasification process above the surface of ground.

Sustainable Development

TYPES OF GASIFIERS

Fixed Bed Gasifiers
Flow of solids is independent of the flow of gas.
Requires minimal pre-treatment of feed coal.
Have high thermal efficiency.

Fluidized Bed Gasifiers
Characterised by uniform particle mixing, uniform temperature gradients with high char recycling rate and less capital cost.

Entrained Flow Gasifiers
Ability to handle practically any feed. Syngas obtained is free of oils and tars.

Clarity for UPSC by Dr. Shivin

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Advantages:

- **Clean** coal technology compared to burning of coal as CG plants do not produce any scrubber sludge;
- Most part of the **wash water is recycled** and the residual waste waters from gasification plants can be effectively treated.
    - Coal can be used to produce number of products using clean coal technologies
    - **Hydrogen, Methanol and Fertilizers** through coal gasification.
    - Carbon Fibres, Plastic composites using ash/ residue of the coal power plant/ gasifier.

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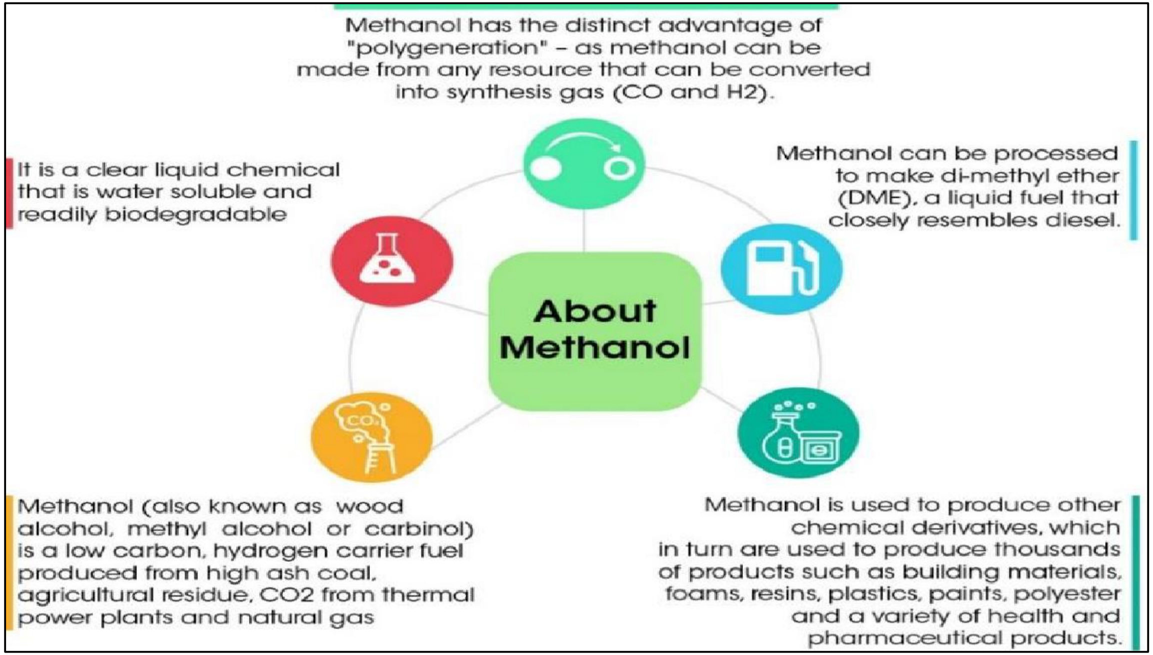
National Coal Gasification Mission initiative

- Under Atma Nirbhar Bharat Abhiyan, **Ministry of Coal** has taken National Coal Gasification Mission initiative for utilizing coal through coal gasification and achieve 100 MT coal gasification by year 2030.
- **Nodal officer:** All coal companies have been advised to appoint a nodal officer and to prepare an action plan for **gasifying at least 10% of their coal production.**
- **Interest Rate Subvention:** Interest Rate Subvention may be provided thereby reducing the interest burden on coal gasification projects and improve the bankability of these projects.
- **Import duty exemption:** Import duty exemption for capital goods for setting up coal gasification projects may be considered.

- **Methanol Energy and Methanol Economy**



Renewable Energy

- Methanol burns efficiently in all internal combustion engines, produces **no particulate matter, no soot, almost nil SOx and Nox emissions.**
- **Blending of 15% methanol** in petrol will reduce pollution by 33% & diesel replacement by methanol will reduce pollution by more than 80%.
- Although slightly lower in energy content than petrol and diesel, methanol can replace both these fuels in Transport sector, Energy sector, Retail cooking etc.
- Methanol & Di-methyl Ether are substantially **cheaper** than Petrol and Diesel.
- Methanol has many desirable attributes which make it an excellent spark ignition engine fuel, including high octane number and improved efficiency.

Renewable Energy

- Methanol Economy is the "Bridge" to the dream of a complete "Hydrogen based fuel systems".
- The Concept of **"Methanol Economy"** is being actively pursued by China (10% of fuel in transport sector).
- Methanol economy will help India use its vast coal reserves (burning coal directly is bad).

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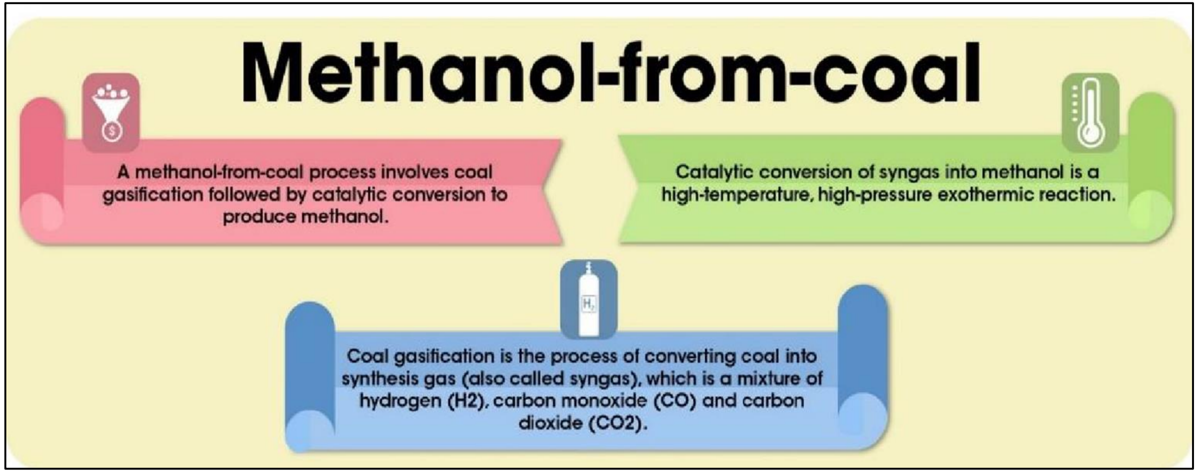
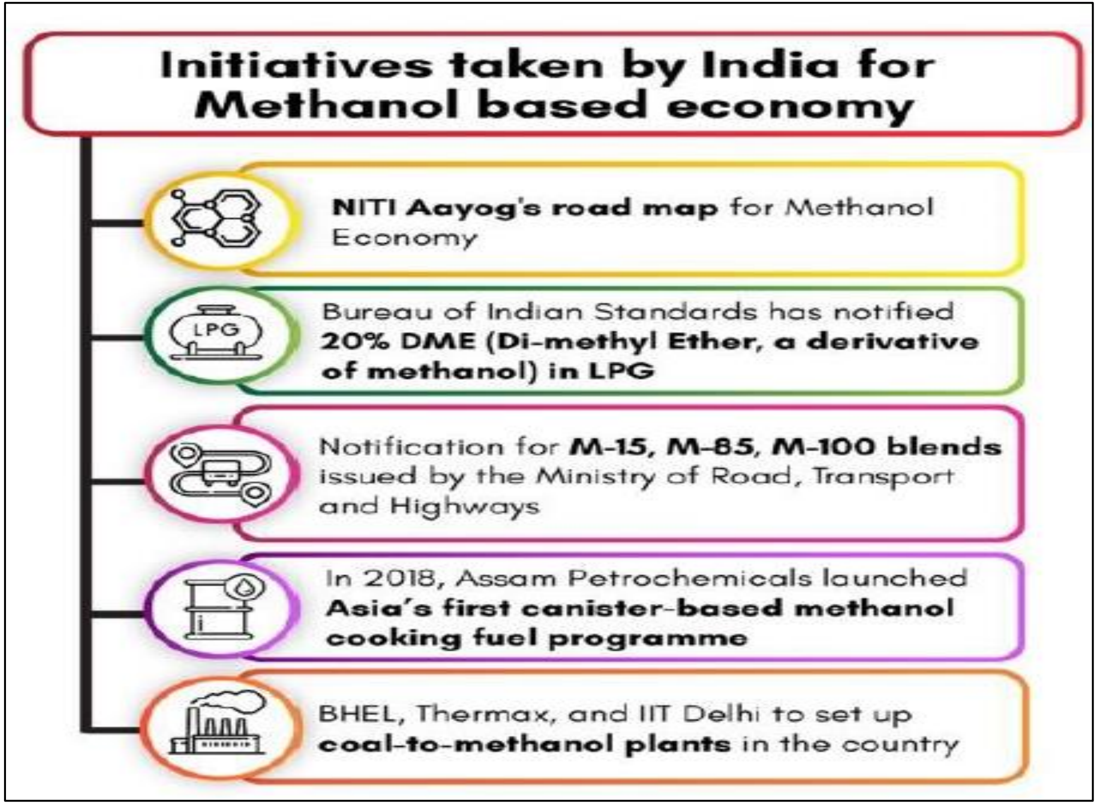
- **<u>Recent initiatives</u>**
- Recently, India's first Indigenously Designed **High Ash Coal Gasification Based Methanol Production Plant** was inaugurated at BHEL R&D Centre, Hyderabad.



Renewable Energy



Methanol	Ethanol
Methyl Alcohol (CH_3OH)	Ethyl Alcohol (**drinking alcohol**) (CH_3-CH_2-OH)
Methanol is toxic and hence it is **not suitable for consumption**.	Ethanol is used in alcoholic beverages, as additive in foods, etc.
Methanol is mostly used to create other chemicals such as **acetic acid** and **formaldehyde**.	It has antiseptic properties and is found in anti-bacterial wipes and hand gels. It is also the base of many paints and perfumes because it is a good solvent.
Its odour is distinctive, and it burns as a bright white flame.	It has a strong, burning odour and will burn as a bright blue flame.
Most of the methanol is produced from **coal** and **natural gas**.	Most of the ethanol is produced by **fermentation of biowaste and food crops**.
• They are both used as solvents. Both are **clean fuels** • **Both can be blended with petrol for automobile fuel**.	

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- **<u>Green Methanol</u>**
- **Context-** National Thermal Power Corporation Limited (NTPC) and Tecnimont (Italy based company) signed MoU to jointly explore green methanol production facility at a NTPC project in India.
- Green Methanol is methanol produced via <u>a process that emits zero or a minimal amount of GHGs,</u> usually measured as an equivalent amount of CO2.
- It is produced from **sustainable biomass**, or from carbon dioxide and hydrogen produced from renewable electricity.
- **Applications**: material for the chemical industry, storing renewable electricity, transportation fuel and substitute fuel for maritime fuel applications

Renewable Energy

- **Methanol blended Diesel (MD15)**
- **Context:** Demo-run of Vessel powered by Methanol blended Diesel (MD15) named SB Gangadhar in Guwahati, Assam was held in run-up to India Energy Week 2023 (IEW 2023).
- IEW 2023, first major event under **India's G20 Presidency,** follows PM's pledge at COP26 to cut India's emissions to net-zero by 2070.
- IEW 2023 is held under **Ministry of Petroleum & Natural Gas.**

Renewable Energy

- <u>**Global International Biofuel Alliance (GIBA)**</u>
- GIBA was launched by **India during India Energy Week 2023.**
- GIBA will help create a <u>favourable ecosystem for promoting development and deployment of biofuels with key stakeholders</u> including US, Brazil, EU, IEA etc. as a clean mobility alternative under India's G20 Presidency.

- **<u>Other Initiatives in news</u>**
- **<u>Network for Greening the Financial System</u>**
- Reserve Bank of India (RBI) has joined the Central Banks and Supervisors Network for Greening the Financial System (NGFS) as a member.
- **About NGFS:**
- The NGFS is a <u>group of Central banks and supervisors willing to share the best practices</u> and contribute to the development of the environment and climate risk management in the financial sector.
- The System was launched at the **Paris One Planet Summit** in December 2017.

Renewable Energy

- **<u>World's First compressed natural gas (CNG) Terminal</u>**
- Prime Minister laid foundation stone of the World's First CNG Terminal and of the brownfield port at **Bhavnagar, Gujarat.**
- It is expected to be operational by **2026.**
- Developed using a <u>public-private partnership (PPP) approach</u> and has a cargo handling capacity of 1.5 million metric tonnes per annum (MMTPA).
- The upcoming project will have direct door-step connectivity to the existing roadway and railway network connecting to the largest industrial zones, Dedicated Freight Corridor and northern hinterland of India.

Renewable Energy

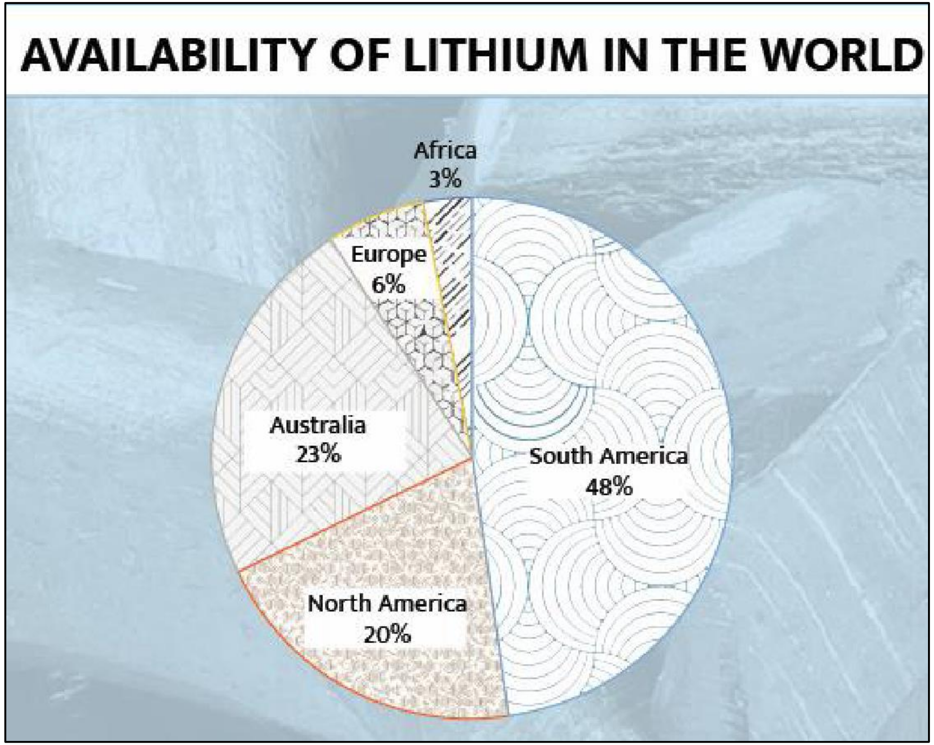
- **Environmental Information, Awareness, Capacity Building and Livelihood Programme (EIACP)**
- **Environmental Information System (ENVIS)** is renamed as EIACP.
- EIACP serves as a one stop platform for dissemination of environmental information, informed policy formulation on various facets of environment and facilitation of alternate livelihoods through green skilling.
- EIACP conducts **Environmental Awareness Programs** to School Children, Colleges and Universities on Environmental Important Days.
- As per the mandate of EIACP, the activities of the Programme Centres are to be aligned with **Lifestyle for the Environment (LiFE)** introduced by India at COP 26 in Glasgow.

Renewable Energy

- **Lithium-ion battery**
- **About Lithium**
- Lithium is **lightest solid metal**. It is currently produced from **hard rock** or brine mines.
- **Australia** is the world's biggest supplier, with production of Lithium from hard rock mines, while Argentina, Chile and China are producing it from salt lakes.
- **Lithium Triangle : Argentina, Bolivia, Chile**

- **Lithium in India**
- In India first traces of Lithium ever to be discovered in the ancient igneous rock of Karnataka's **Mandya** district.
- Recently Lithium reserves were also discovered in J&K. (please read up from geography about minerals prospecting etc.)
- Currently India **imports** all its lithium needs, with **China** and **Hong Kong** biggest suppliers.

Renewable Energy

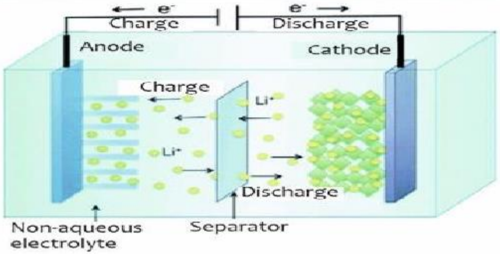


Lithium-Ion batteries

A lithium-ion battery is a **family of rechargeable battery types**and consists **four components** in a lithium-ion cell: **anode, cathode,separator,** and **the aqueous electrolyte.**

Anode	**Cathode**
Graphite	$LiCoO_2$
Separator	**Aqueous electrolyte**
Polypropylene (PP) and and Polyethylene (PE)	organic solvent and dissolved lithium salt

Lithium-ion Battery



Advantages of Li-ion batteries

- **High specific charge density** (amount of charge a battery can supply)
- **High Voltage**
- **High energy density**
- **Low self-discharge**

Disadvantages of Li-ion batteries

- **Thermal runaway** (due to melting of the electrolyte, high operational temperature)
- **Less stability**
- **Lack of availability** for lithium raw material

Renewable Energy

- **Steps taken for Lithium-ion batteries in India**
- **National Mission on Transformative Mobility:** To encourage domestic Lithium- Ion Cell manufacturing and EV components.
- ISRO and BHEL agreement: to develop low-cost lithium-ion batteries.
- India's **first lithium cell plant manufacturing** facility will be launched in **Tirupati**, Andhra Pradesh.
- Lithium triangle nations: India is focusing on **'Lithium Triangle'** nations Argentina, Bolivia and Chile for joint exploration

Comparison of various battery types

Type	Advantage	Disadvantage
Lead Acid batteries	**Mature technology, low cost** and ruggedness compared with other battery technologies.	**Heavy and bulky**, do not cycle well.
Nickel-cadmium batteries	**Availability in all sizes**, can be moved around easily, can be **stored, or used for up to 5 years, offer more than 1,000 charge and discharge cycles.**	**Environmental concerns** related to cadmium. **Prone to memory effect** that can cause a battery to lose electrical storage area. **Higher self-discharge rate** than Lithium-ion.
Nickel metal hydride batteries	Widely used in **consumer electronics**, addresses the issues arising from the environmental effects of cadmium.	**Not considered for large stationary applications because of high cost** of Nickel, also have a high self-discharge rate and generally take a long time to charge.
All solid-state batteries	**High energy densities** and charge/discharge current limits, a higher life even in harsh conditions, lower charging time.	Challenges in both **manufacturing and fundamental technology understanding.**
Metal air	**High energy density** than lithium-ion batteries, only anode replacement needed, comparatively longer shelf life than lithium-ion.	**Not electrically rechargeable,** lower charge/discharge rates compared with Lithium-ion batteries
Sodium ion batteries	Wider operational temperature range and are safer. Cheaper than Lithium-ion.	Lower energy density than lithium-ion batteries.

Reports in News

Report	Published by
Fostering Effective Energy Transition	World Economic Forum (WEF)
Environment Performance Index (EPI)	World Economic Forum (WEF) in collaboration with Yale Center for Environmental Law and Policy and Columbia University Center for International Earth Science Information Network
Renewables 2022 Global Status Report	REN21 (only global community of actors from science, governments, NGOs and industry working collectively to drive the rapid uptake of renewables)
Renewable Capacity Statistics 2022	International Renewable Energy Agency (IRENA)

Reports in News

Report	Published by
Renewable Power Generation Costs in 2021	IRENA
Renewable Energy (RE) and Jobs Annual Review 2022	IRENA and the International Labour Organization
Renewable Energy (RE) Report 2022	International Energy Association (IEA)
World Energy Employment Report 2022	International Energy Agency (IEA)
World Energy Outlook 2022	International Energy Agency (IEA) (Annual report)
Global Wind Report 2022	Global Wind Energy Council

Reports in News

Report	Published by
State of India's Environment (SoE) in figures, 2022	Centre for Science and Environment
EnviStats India 2022	National Statistics Office (NSO)
Renewables Integration in India 2021	NITI Aayog and International Energy Agency (IEA)
Transforming Food Systems for Rural Prosperity	International Fund for Agricultural Development (IFAD)
A Multi-Billion-Dollar Opportunity: Repurposing Agricultural Support to Transform Food System	FAO, UNEP, UNDP

- **<u>Stubble Burning</u>**
- Stubble burning is the act of **setting fire** to the **crop** residue that remains after grains like paddy, wheat etc. have been harvested in order to prepare field for sowing next crop.
- It is one of the chief causes for rising air pollution levels in **Delhi** during winters.
- **<u>Reasons for Stubble burning</u>**
- **Mechanization** (use of combine harvesters) leaves behind smaller stubble on the ground that is difficult to collect.

Pollution

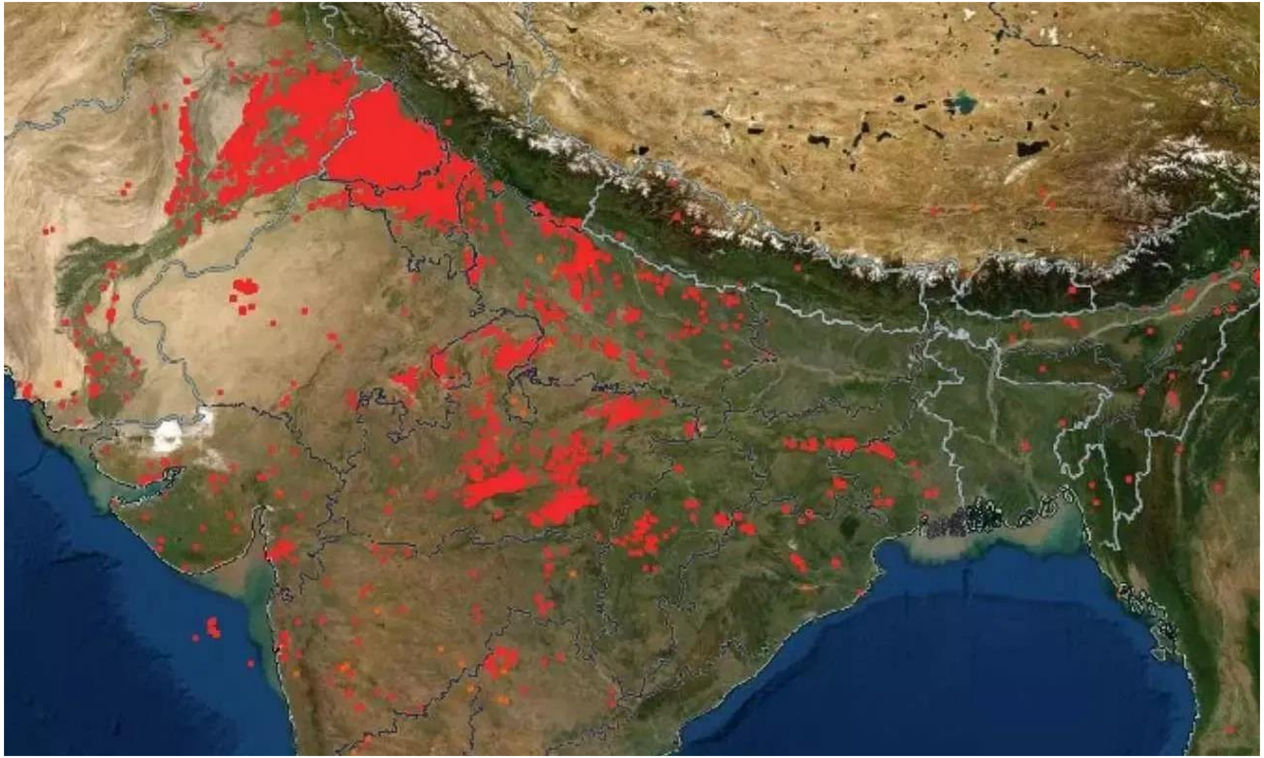
- Laws like **Punjab Preservation of Subsoil Water Act** which has delayed date for paddy transplantation.
    - The Act aims at **conserving groundwater** by mandatorily delaying the transplanting of paddy to beyond June 10, when the most severe phase of **evapotranspiration** (transfer of water from land to the atmosphere through evaporation from the soil and plant transpiration) is over.
    - Farmers were forbidden from **sowing paddy before May 10**, and transplanting it before June 10.
    - Farmers' organisations say late sowing and transplanting **delays the harvesting** as well (it is end-October by the time operations end), and they are left with a very small window to prepare their fields for the next (wheat) crop.
    - In this situation, setting the stubble ablaze is a quick-fix solution.

- **Crop intensification** leaves farmers with small windows to clear the fields so burning the residue is the easiest.
- **<u>Measures to control Stubble burning</u>**
- **In-situ Measures:** Use of **Happy seeder Machines**; Custom Hiring Centers to make this equipment available to small and marginal farmers on reasonable rent; Promotion of short duration rice varieties and crop diversification etc
  - Happy Seeder Machine is a machine **mounted on a tractor** that not only **cuts and uproots the stubble**, but can also drill wheat seeds on the soil that has just been cleared up.
  - The straw is simultaneously thrown over the sown seeds to form a mulch cover.
- **Ex-situ measures:** Using stubble in different ways like cattle feed, compost manure, roofing in rural areas, **biomass energy**, packing materials etc.

Pollution

- **Policy response to address stubble burning**
- **National Green Tribunal** had **banned** crop residue **burning** in the states of Rajasthan, Uttar Pradesh, Haryana and Punjab.
- **National Policy for Management of Crop Residue (NPMCR),** 2014 to ensure prevention of burning of crop residues, by incentivizing purchase of modern machineries.
- Central Sector **Scheme** on ‘Promotion of Agricultural Mechanization for In-Situ Management of Crop Residue in the States of Punjab, Haryana, Uttar Pradesh and NCT of Delhi’.
- Supreme Court guideline to incentivise farmers for stopping the burning of paddy crop stubble.
- Commission for Air Quality Management directions

Pollution



- **PUSA Bio-Decomposer**
- Developed by Indian Council of Agricultural Research (**ICAR**) , it is a low-cost **microbial bio-enzyme** solution which accelerates the decomposition of crop residue, including stubble from paddy crops into manure within **15- 20 days.**
- In one capsule of **fungal solution**, a farmer can make 25 litres of liquid substance. After making the liquid, he has to add jaggery and gram flour in it and has to sprinkle that liquid on the stubble, after which it would bio-degrade in 20 days.
- It is emerging as a potent solution towards curbing stubble burning.
- Also, it provides **additional manure** through decomposed stubble, improving the soil quality as well.

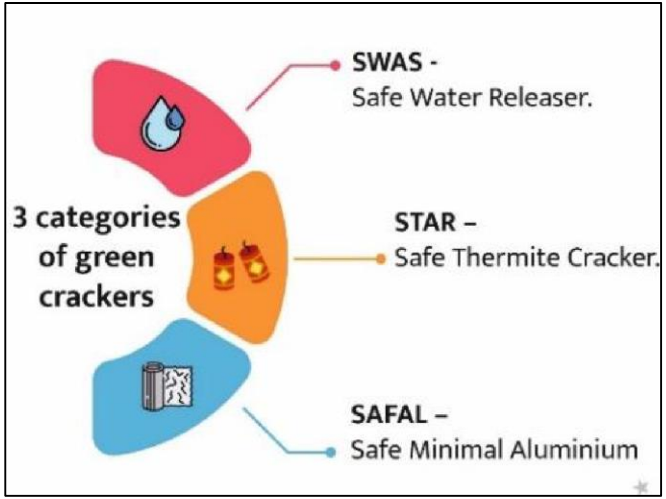
Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

- **<u>Anti-smog gun</u>**
- It was recently installed in **Delhi** to reduce air-pollution mainly concentration of **PM10 and PM2.5.**
- Anti-smog gun is designed to create an ultra-fine fog, comprising very fine water droplets (less than 10-micron in size).
- These tiny water droplets will be spread over a sizeable area with the help of a high-speed fan, which can absorb smallest dust particles in air.

Pollution

- <u>**Green crackers**</u>
- SC, in **2018** judgement on firecrackers banned manufacture, sale and use of joined firecrackers (series crackers or ‘laris’) as they caused air, noise and solid waste problems.
- SC allowed the manufacture and sale of **only “green”** (safe water and air sprinklers that emit less sound and light) or “improved” crackers (avoid the use of ash as filler material).



- **<u>About Green Crackers</u>**
- They are made using **less polluting raw materials**. Their chemical formulation ensures reduced particle emission into the atmosphere by suppressing the dust produced. They do not contain hazardous chemicals like **Barium Nitrate.**
- Green crackers were researched and developed by **CSIR-NEERI.**
- Different types of green crackers available in India: Safe Water Releaser (**SWAS**), Safe Thermite Cracker (**STAR**) and Safe Minimal Aluminium (**SAFAL**) crackers.
- Petroleum and Explosives Safety Organisation (PESO) under Ministry of Commerce and Industry tasked with certifying that crackers are made without arsenic, mercury, and barium, and are not loud beyond a certain threshold

Pollution

- **Commission for Air Quality Management (CAQM)**
- A **statutory body** established under the Commission for Air Quality Management in National Capital Region and Adjoining Areas Act, 2021.
- First established in 2020 through an Ordinance.
- Headed by **Chairperson** with at least 15 years' experience in the field of environment and pollution or 25 years of administrative experience.
- Appeal of CAQM orders, directions etc.**: National Green Tribunal** (recall what other Acts does NGT have jurisdiction to adjudicate upon)

Pollution

- **Functions of CAQM**
- **Coordinate** actions on monitoring of air quality.
- **Planning** and executing plans to prevent and control air pollution in NCR.
- Conduct **research** and development through networking with technical institutions.
- **Recent news**
- The CAQM issued directions to **ban the use of coal** in industrial, domestic and other miscellaneous applications to bring down GHG emission.
- Use of **low-sulphur coal** in thermal power plants has been exempted from the ban.

Pollution

- Date of implementation of ban on use of coal
- October 1, 2022 in areas having PNG (piped natural gas) infrastructure and supply.
- January 1, 2023 in areas where PNG supply is still not available.
- **Graded Response Action Plan (GRAP)**
- CAQM issued orders for enforcing measures under ‘stage-1’ of **GRAP** in National Capital Region (NCR) after air quality deteriorated to be in ‘poor’ category in Delhi.
- GRAP: An **action plan** to address **air pollution** in Delhi-NCR, based on 4 different stages of adverse air quality

Pollution

- It was revised recently by CAQM under provisions of CAQM in NCR and Adjoining Areas Act, 2021.

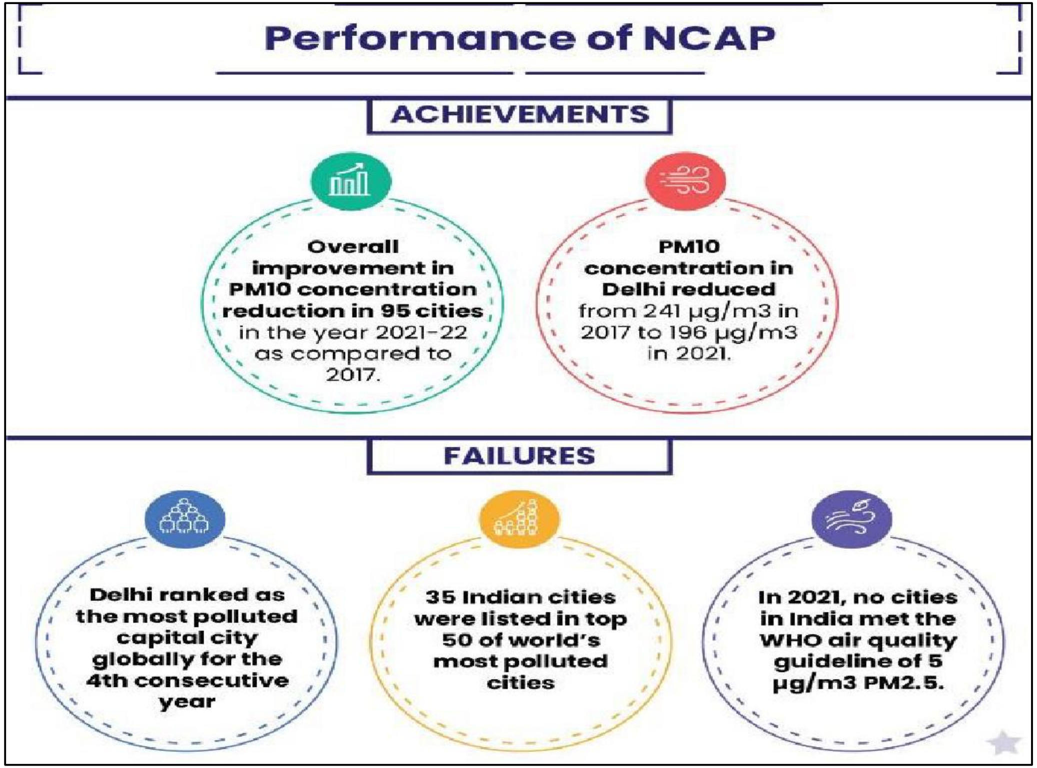
Graded Response Action Plan (GRAP)	
Air Quality (AQ)	Prescribed Measures
Stage I - 'Poor' (201-300)	• **Prohibition on construction and demolition activities** at plots equal to or more than 500 sqm. Spreading **public awareness** for reporting air pollution incidents etc.
Stage II – 'Very poor' (301- 400)	• Ensuring **uninterrupted power supply** to discourage use of generator sets. **Ban on use of diesel generator** sets except for emergent and essential services including railway etc.
Stage III – 'Severe' (401-450)	• **Closure of brick kilns and hot mix plants** not operating on fuels approved by CAQM.
Stage IV – 'Severe+'	• **Four-wheeler diesel light motor vehicles will not be allowed.** **Closure of all industries in NCR not running on fuels approved** by CAQM.

Pollution

- **<u>National Clean Air Program</u>**
- Launched in **2019** for reducing for comprehensive mitigation actions for prevention, control and abatement of air pollution.
- Initially envisaged to achieve targets of **20 to 30 %** reduction in Particulate Matter (PM10 and PM2.5) concentration by **2024** across the country taking 2017 as the base year.
- **<u>Recent changes</u>**
- New target of a **40%** reduction in particulate matter concentration in cities covered under the Programme by **2026**

- **Key features**
- Preparation of **City-specific action plans** which include measures for strengthening the air quality monitoring network, reducing vehicular and industrial emissions, increasing public awareness, etc.
- Implemented in **132 cities-**
- **123 Non-Attainment cities (NACs)** identified under NCAP based on non-conforming to National Ambient Air Quality Standards (NAAQS) consecutively for five years.
- **8 million Plus Cities (MPCs)** identified by 15th Finance Commission (XV-FC), for receiving performance-based grant for air quality improvement.
- **PRANA** (Portal for Regulation of Air-pollution in Non-Attainment cities) portal monitors the implementation of NCAP.

Pollution


Performance of NCAP
ACHIEVEMENTS
Overall improvement in PM10 concentration reduction in 95 cities in the year 2021-22 as compared to 2017.
PM10 concentration in Delhi reduced from 241 µg/m3 in 2017 to 196 µg/m3 in 2021.
FAILURES
Delhi ranked as the most polluted capital city globally for the 4th consecutive year
35 Indian cities were listed in top 50 of world's most polluted cities
In 2021, no cities in India met the WHO air quality guideline of 5 µg/m3 PM2.5.

Pollution

- **<u>Swachh Vayu Sarvekshan</u>**
- MoEFCC will launch Swachh Vayu Sarvekshan under NCAP.
- 131 cities to be ranked in the country for implementing City Action Plans prepared NCAP for reducing air pollution upto 40% by 2025-26.
- Cities categorized into 3 groups based on population (population > 10 lakh, population between 3 to 10 lakh and population < 3 lakh).
- For the survekshan cities are required to do the **annual self-assessment** as per the framework provided on **PRANA online portal.**

Pollution

- **Report implementation** of activities and measures taken in respect of solid waste management, road dust management, management of construction and demolition waste, control of vehicular emissions and industrial pollution.
- Based on the self-assessment and third party assessment, **3 best performing cities in each group will be given cash award.**

Pollution

- **National Ambient Air Quality Standards (NAAQS)**
- NAAQS is developed by the Central Pollution Control Board (CPCB).
- They are applicable all over the country.
- The CPCB has been conferred this power by the Air (Prevention and Control of Pollution) Act, 1981.

Pollution

National Ambient Air Quality Standards, as of 2009			
Pollutants (12)	Time Weighted Average	Concentration in Ambient Air	
		Industrial, Residential, Rural & Other Area	Ecologically Sensitive Area (notified by GOI)
SO_2, µg/m^3	Annual	50	20
	24 hours	80	80
NO_2, µg/m^3	Annual	40	30
	24 hours	80	80
PM_{10}, µg/m^3	Annual	60	60

Pollution

PM_{10}, µg/m³	24 hours	100	100
$PM_{2.5}$, µg/m³	Annual	40	40
	24 hours	60	60
O_3, µg/m³	8 hours	100	100
	1 hour	180	180
Lead (Pb), µg/m³	Annual	0.50	0.50
	24 hours	1	1
CO, mg/m³	8 hours	2	2
	1 hour	4	4
Ammonia (NH_3), µg/m³	Annual	100	100
	24 hours	400	400
Benzene, µg/m³	Annual	5	5
Benzopyrene, µg/m³	Annual	1	1
Arsenic (As), ng/m³	Annual	6	6
Nickel (Ni), ng/m³	Annual	20	20

Pollution

P	**PM 2.5**
M	**PM 10**
publish	**Lead**
O	**Ozone**
N	**NO_2**
N	**NH_3 : ammonia**
S	**SO_2**
C	**CO**
B	**Benzene**
B	**Benzopyrene**
A	**Arsenic**
N	**Nickel**

- **<u>WHO's Air Quality Database</u>**
- Context- WHO has released **2022 update** of its air quality database.
- **<u>About the WHO Air quality guidelines</u>**
- Recommended levels and interim targets for common air pollutants: **PM, O3, NO2, and SO2.**
- First released in **1987** and Latest global version published in 2021.
- **<u>About the 2022 update</u>**
- Almost the entire global population (99%) breathes air that exceeds WHO air quality limits.

Pollution

- Introduces for the first time, **ground measurements** of annual mean concentrations of nitrogen dioxide (**NO2**), a common urban pollutant and precursor of particulate matter and ozone.
- Includes measurements of particulate matter with diameters equal or smaller than 10 µm (PM10) or 2.5 µm (PM2.5).

WHO's 2021 Air Quality Guidelines

Pollutant	Averaging Time	2021 AQGs
$PM_{2.5}$, µg/m^3	Annual	5
	24-hour[a]	15
PM_{10}, µg/m^3	Annual	15
	24-hour[a]	45
O_3, µg/m^3	Peak season[b]	60
	8-hour[a]	100
NO_2, µg/m^3	Annual	10
	24-hour[a]	25
SO_2, µg/m^3	24-hour[a]	40
CO, mg/m3	24-hour[a]	4

- <u>**National Air Quality Monitoring Programme**</u>
- The Central Pollution Control Board (CPCB) has been executing a nationwide programme of ambient air quality monitoring known as National Air Quality Monitoring Programme (NAMP).
- The network consists of **804 operating stations** covering 344 cities/towns in 28 states and 6 UTs of the country.
- NAMP is undertaken to determine the status and trends of ambient air quality; to ascertain the compliance of NAAQS; to identify **non-attainment cities**; to understand the natural process of cleaning in the atmosphere; and to undertake preventive and corrective measures.

Pollution

- Under NAMP, **four** air pollutants

1. Sulphur Dioxide (**SO2**),
2. Oxides of Nitrogen (**NO2**),
3. Respirable Suspended Particulate Matter (**RSPM / PM10**)
4. Fine Particulate Matter (**PM2.5**)

- have been identified for regular monitoring at all the locations.
- The monitoring **wind speed** and **wind direction**, relative humidity (RH) and **temperature** were also integrated with the monitoring of air quality.

- **System of Air Quality and Weather Forecasting and Research (SAFAR)**
- It is a national initiative introduced by the **Ministry of Earth Sciences** (MoES) (**not** MoEFCC).
- It is an integral part of India’s first Air Quality Early Warning System operational in Delhi.
- SAFAR will measure the air quality of a metropolitan city.
- It also monitors all weather parameters like temperature, rainfall, humidity, wind speed, and wind direction, UV radiation, and solar radiation.
- The network has **multiple air quality monitoring stations** in each city, equipped with real-time, continuous monitors for various pollutants)

Pollution

- The system is indigenously developed by the **Indian Institute of Tropical Meteorology (IITM), Pune** and is operationalised by the India Meteorological Department (IMD).
- It has a giant colour LED display that gives out real-time air quality index on a 24x7 basis with **colour-coding** (along with 72 hours advance forecast).
- The ultimate objective of the project is to increase awareness among the public regarding the air quality in their city.

Pollution

- Pollutants monitored under SAFAR System include:

1. PM2.5,
2. PM10,
3. Ozone,
4. Carbon Monoxide (CO),
5. Nitrogen Oxides (NOx),
6. Sulphur Dioxide (SO2),
7. Mercury
8. Benzene
9. Toluene (methylbenzene — used in paint thinners & as octane booster in gasoline engines),
10. Xylene (dimethylbenzene — used as a solvent in printing, rubber, and leather industry), and

Pollution

Degrees of Air Pollution			
SAFAR AQI Scale		**World AQI Scale**	
Good	0-50	Good	0-50
Satisfactory	51-100	Satisfactory	50-100
Moderate	100-200	Unhealthy**	100-150
Poor	201-300	Unhealthy	150-200
Very Poor	301-400	Very Unhealthy	200-300
Severe	401-500	Severe	Over 300

Pollution

- **National Air Quality Index (AQI)**
- AQI was launched by **MoEFCC** in April 2015 under 'Swachh Bharat'.
- It helps the citizens to judge the air quality within her vicinity.
- There are **six AQI** categories, namely Good, Satisfactory, Moderately polluted, Poor, Very Poor, and Severe.
- The AQI will consider **eight** pollutants
- PM10 and PM2.5
- NO2, SO2,
- Carbon monoxide (CO), Ozone (O3)
- Ammonia (NH3) and Lead (Pb)

- **<u>National Air Quality Resource Framework of India (NARFI)</u>**
- NARFI is an **information mechanism** to help decision-makers in government, municipalities, start-ups and in private sectors to address air pollution issues in different climatic zones of India.
- Developed by**: National Institute of Advanced Studies (NIAS)**, Bengaluru with support from Office of Principal Scientific Adviser.
- Will provide an all-inclusive guide to collecting air quality data, studying its impact and implementing science based solutions.
- Will have **5 modules** including Emission Inventory, Air Shed, and Mitigation; Impacts on Human Health and Agriculture etc.

- <u>**Norms for thermal power plants (TPPs)**</u>
- Context: Ministry of Environment, Forest and Climate Change (MoEFCC) gave two year extension to thermal power plants (TPPs) on SO2 norms.
- <u>**About TPP Norms**</u>
- In 2015, MoEFCC notified environmental norms for particulate matter (**PM**), sulphur dioxides (**SO2**), **nitrogen oxides** and **mercury** and **water use** for coal based TPPs.
- Initially deadline for implementation- 2017 was extended to 2022.

Coal TPPs **contribute to over half SO_2 concentration, 30% oxides of nitrogen, 20% PM** in the ambient air..

Pollution

- Deadline for SO_2 further extended for all three categories of TPPs.
- However, deadline remains **same** for meeting **PM and NOx standards.**
- Compliance with norms requires retrofitting existing TPPs with auxiliaries to control emissions such as **Flue Gas Desulphurisation (FGD),** Selective Catalytic Reduction etc.
- FGD: A set of technologies used to remove SO_2 from sources of emissions, e.g., exhaust flue gases of fossil-fuel power plants.
- **Thermal power plants** are the **biggest users** of water among all industries.

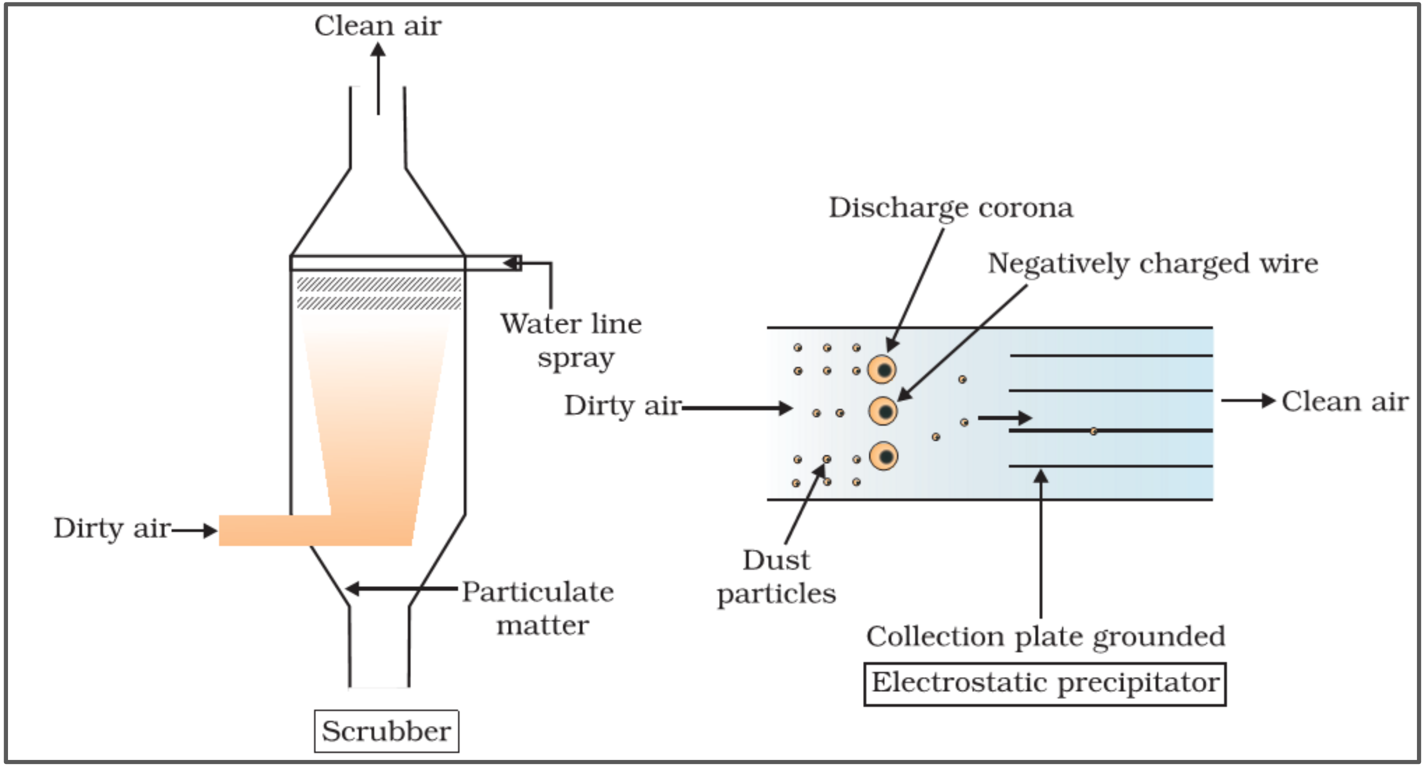
Pollution

SO_2 Norms for Thermal Power Plants (TPPs)			
Category	Criteria	Deadline for parameters other than SO_2 emissions	Deadline for SO_2 emissions
A	Within 10 km radius of the National Capital Region (NCR) or cities having million-plus population .	End of 2022	End of 2024
B	Within 10 km radius of critically polluted areas or non-attainment cities (not meeting National Ambient Air Quality Standards).	End of 2023	End of 2025
C	Remaining plants	End of 2024	End of 2026

- **Electrostatic Precipitators (ESP)**
- Electrostatic precipitation can remove over **99 per cent of particulate matter** present in the exhaust.
- They are used in **boilers** and **furnaces**, thermal power plants, cement factories, steel plants, etc.
- In electrostatic precipitation, emanating dust is **charged with ions**, and the ionised particulate matter is collected at an oppositely charged surface.

- **<u>Working</u>**
- An electrostatic precipitator has **electrode wires** that are maintained at several thousand volts, which produce a corona that releases electrons.
- These electrons attach to dust particles giving them a **net negative charge**.
- The collecting plates are **grounded** (relatively positive charge) and attract the charged dust particles.
- The air velocity between the plates must be low enough to allow the dust to fall.
- The particles are removed from the collection surface by **occasional shaking or rapping**.

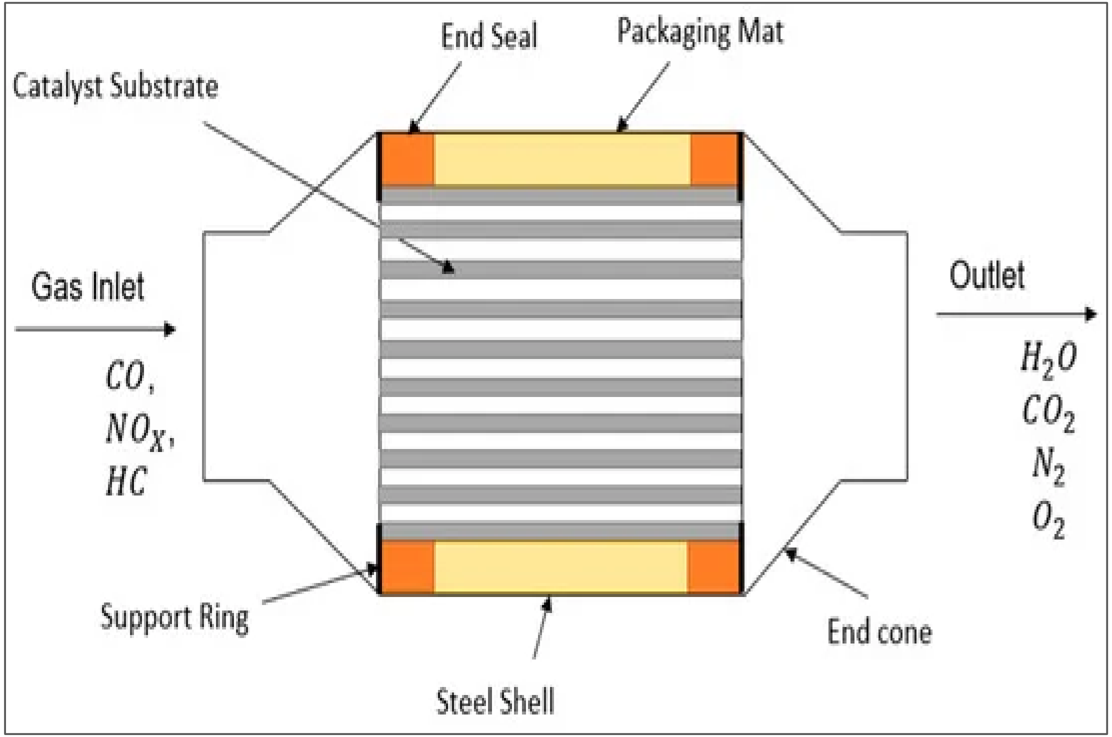
Pollution



Clean air
Water line spray
Dirty air
Particulate matter
Scrubber
Discharge corona
Negatively charged wire
Dirty air
Clean air
Dust particles
Collection plate grounded
Electrostatic precipitator

- **<u>Scrubbers</u>**
- Scrubbers are **wet collectors**. They remove aerosols from a stream of gas either by collecting wet particles on a surface, or the particles are wetted by a scrubbing liquid.
- The particles get **trapped** as they travel from the supporting gaseous medium across the interface to the liquid scrubbing medium (this is similar to mucus in trachea trapping dust).
- A scrubber can remove gases like **sulphur dioxide**.
- **<u>Inertial Collectors</u>**
- Inertia of **Suspended Particulate Matter** in gas is higher than its solvent. As inertia is a function of the mass of the particulate matter, the device collects heavier particles more efficiently (centrifugation is the technique).
- **‘Cyclone’** is a common inertial collector used in gas cleaning plants.

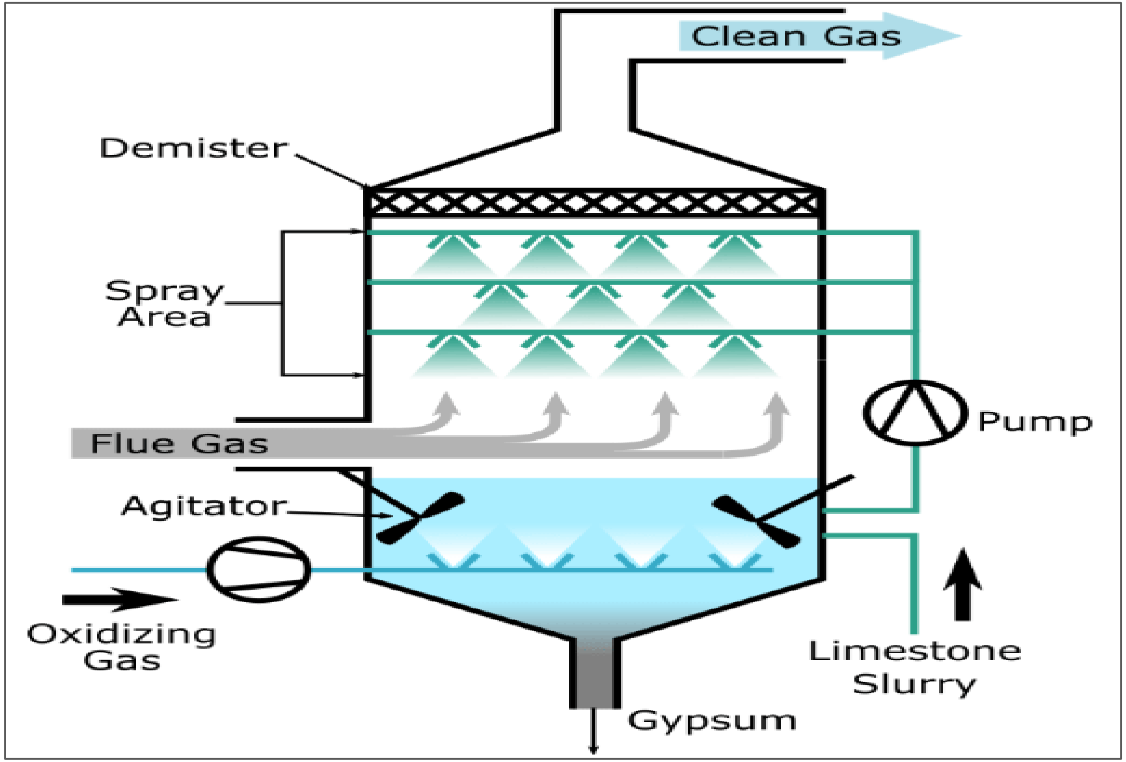
- **<u>Catalytic Converter</u>**
- Catalytic converters, having expensive metals, **platinum-palladium** and **rhodium** as the catalysts, are fitted into automobiles for reducing the emission of poisonous gases.
- Motor vehicles equipped with catalytic converters should use **unleaded petrol** because the lead in the petrol inactivates the catalyst.
- **<u>Working</u>**
- As the exhaust passes through the catalytic converter, **unburnt hydrocarbons** are converted into **carbon dioxide and water**, and carbon monoxide and nitric oxide are changed to carbon dioxide and nitrogen gas, respectively.



End Seal
Packaging Mat
Catalyst Substrate
Gas Inlet
CO,
NO_X,
HC
Outlet
H_2O
CO_2
N_2
O_2
Support Ring
End cone
Steel Shell

- **Flue-Gas Desulfurization (FGD) systems**
- FGD systems may involve wet scrubbing or dry scrubbing to remove acidic gases, particularly sulphur dioxide (SO2) and HCl, from flue gases (gases flowing out of vertical exhaust pipes). **Lime** is used in both.
- **Wet FGD**
  - Flue gases are brought in contact with a **liquid or solid slurry absorbent**.
  - The sulphur dioxide dissolves in or reacts with the absorbent & becomes trapped in as wet calcium sulphite.
  - The sulphite can be converted to **gypsum** (used in cement production).
- In **dry FGD** scrubbing, lime is injected directly into flue gas to remove SO2 and HCl from low-sulphur fuels.

Pollution


Clean Gas
Demister
Spray
Area
Flue Gas
Pump
Agitator
Oxidizing
Gas
Limestone
Slurry
Gypsum

- **<u>Fly ash</u>**
- Fly ash is a by-product of **coal based power generation**.
- It is a fine powder with substantial amounts of oxides of silica, aluminium and calcium.
- It also contains traces of **Arsenic, Boron, Chromium**, lead etc. which leads to air and water pollution if disposed on land.
- With **low grade of Indian coal**, its ash content is as **high** as 30-45% in comparison to imported coal with 10-15%.
- With nearly 55% of our total power production through coal and lignite based Thermal Power Plants (TPP), the fly ash generation in India is very high.
- About 83% of Fly Ash is utilized.

Pollution

- **<u>Recent news</u>**
- The Ministry of Environment, Forests and Climate Change (MoEFCC) has extended fly ash utilization deadline for thermal power plants with the introduction of penalties for noncompliance.
- **Shorter Fly-ash utilization cycle:** Existing provisions allow TPPs to fully utilize fly ash in a four-year cycle in a staggered manner. The new policy will follow a **three-year cycle** for **100% utilization** of Fly-ash with a grace period of a year if the percentage of ash utilization is between 60-80% and two years if it is below 60%.
- Introduction of **Polluter Pays Principle**: A fine of Rs 1,000 per tonne of unutilized ash has been introduced if the plant does not achieve at least 80% ash utilization annually or in three years.

PYQs: Q1

2020

In India, why are some nuclear reactors kept "IAEA Safeguards" while others are not?

a)Some use uranium and others use thorium

b)Some use imported uranium and others use domestic supplies

c)Some are operated by foreign enterprises and others are operated by domestic enterprises

d)Some are State-owned and others are privately-owned

PYQs: Q1

Answer: B) Some use imported uranium and others use domestic supplies

- There are at present 22 operational reactors, of which 14 are under the International Atomic Energy Agency (IAEA) safeguards as they use imported fuel. India currently imports uranium from Russia, Kazakhstan and Canada. Plans are also afoot to procure the fuel from Uzbekistan and Australia. The remaining "military" facilities remained off-limits to international inspectors.
- By placing the reactors under the IAEA safeguards, India gives the international nuclear energy watchdog access to them.
- This step was taken by the country in 2014 to demonstrate that its nuclear energy programme was for peaceful purposes. This is a necessary step under the Indo-US nuclear deal.

PYQs: Q2

2021

Among the following which one is the least water-efficient crop?

[A] Sugarcane

[B] Sunflower

[C] Pearl millet

[D] Red gram

PYQs: Q2

Answer: a) Sugarcane

- **Option a is correct.** Sugarcane is the least water efficient crop. Among the given options, its water requirement is 1800-2200 mm/season which is highest.
- **Option b is incorrect.** Sunflower's water requirement was estimated at 672.4 mm/season.
- **Option c is incorrect.** Pearl millet is a drought tolerant crop. Water requirement: 350 mm/season.
- **Option d is incorrect.** Red gram uses about 250-400 mm/season of water.

PYQs: Q3

2020

According to India's National Policy on Biofuels, which of the following can be used as raw materials for the production of biofuels?

1) Cassava

2) Damaged wheat grains

3) Groundnut seeds

4) Horse gram

5) Rotten potatoes

6) Sugar beet

Select the correct answer using the code given below:

a) 1, 2, 5 and 6 only

b) 1, 3, 4 and 6 only

c) 2, 3, 4 and 5 only

d) 1, 2, 3, 4, 5 and 6

PYQs: Q3

Answer: a) 1, 2, 5 and 6 only

- Under the National Policy on Biofuels ‘bioethanol’ is defined as ethanol produced from biomass such as sugar-containing materials, like sugar cane, sugar beet, sweet sorghum, etc.; starch-containing materials such as corn, cassava, rotten potatoes, algae, etc.; and, cellulosic materials such as bagasse, wood waste, agricultural and forestry residues or other renewable resources like industrial waste.
- For Ethanol Production the following raw materials may be potentially used: B-Molasses, Sugarcane juice, biomass in form of grasses, agriculture residues (Rice straw, cotton stalk, corn cobs, sawdust, bagasse, etc.), sugar-containing materials like sugar beet, sweet sorghum, etc. and starch containing materials such as corn, cassava, rotten potatoes, etc., Damaged food grains like wheat, broken rice, etc. which are unfit for human consumption, Food grains during the surplus phase. Algal feedstock and the cultivation of seaweeds can also be a potential feedstock for ethanol production. For Biodiesel Production: Nonedible Oilseeds, Used Cooking Oil (UCO), Animal tallow, Acid Oil, Algal feedstock, etc. For Advanced Biofuels: Biomass, MSW, Industrial waste, Plastic waste, etc.

PYQs: Q4

2017

It is possible to produce algae based biofuels, but what is/are the likely limitation(s) of developing countries in promoting this industry?

1. Production of algae based biofuels is possible in seas only and not on continents.

2. Setting up and engineering the algae based biofuel production requires high level of expertise/technology until the construction is completed.

3. Economically viable production necessitates the setting up of large scale facilities which may raise ecological and social concerns.

Select the correct answer using the code given below:

a)1 and 2 only

b)2 and 3 only

c)3 only

d)1, 2 and 3

PYQs: Q4

Answer- B) 2 and 3 only

- Algae can be grown using land and water not suitable for plant and food production. The algal biofuel technology includes selection of specific species for production and extraction of valuable co-products.
- Algae is bioengineered for achieving advanced photosynthetic efficiencies through continued development of production system. To make the industry commercially viable, researchers have pursued biotech varieties, which could be particularly dangerous if released into nature.
- Some algae biodiesel proposals involve aquaculture-style operations in open ocean waters, which could have harmful ecological effects, especially if biotech algae is used. Water and nutrient use would also be extreme, making any serious scaling up of algae biofuels quite unsustainable.

PYQs: Q5

2013

Consider the following organisms:

1. Agaricus
2. Nostoc
3. Spirogyra

Which of the above is/are used as biofertilizer/biofertilizers

a) 1 and 2
b) 2 only
c) 2 and 3
d) 3 only

PYQs: Q5

Answer: B

- Biofertilizers are products containing carrier-based (solid or liquid) living microorganisms which are agriculturally useful in terms of nitrogen fixation, phosphorus solubilization, or nutrient mobilization to increase the productivity of soil or crop.
- Classification of biofertilizers:
  - Bacterial Biofertilizers: Rhizobium, Azospirilium, Azotobacter, Phosphobacteria, Nostoc, etc.
- Agaricus is an **edible fungus** and is commonly known as a mushroom.
- Spirogyra is a **large genus of freshwater** green algae found in shallow ponds, ditches, and amongst vegetation at the edges of large lakes, generally growing free-floating. **Hence, option b is correct**.

PYQs: Q6

2013

With reference to the usefulness of the by-products of the sugar industry, which of the following statements is/are correct?

1. Bagasse can be used as biomass fuel for the generation of energy.
2. Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilizers.
3. Molasses can be used for the production of ethanol.

Select the correct answer using the codes given below.

a) 1 only
b) 2 and 3 only
c) 1 and 3 only
d) 1, 2 and 3

PYQs: Q6

Answer: C

- Bagasse is a remnant left after extracting the juice of the sugarcane or sorghum known as bagasse. It can be used as a fuel for the generation of heat energy and electricity, in the production of pulp, and paper and to feed cattle. **Hence, statement 1 is correct**.

- Molasses is the thick syrup that results from the refining of sugarcane or sugar beets into sugar. It can be used as sweetening and flavoring foods, in the production of commercial brown sugar, baking, ethanol, and cattle feed. **Hence, statement 3 is correct**.

PYQs: Q7

2022

System of Rice Intensification' of cultivation, in which alternate wetting and drying of rice fields is practised, results in :

1. Reduced seed requirement
2. Reduced methane production
3. Reduced electricity consumption

Select the correct answer using the code given below :

[A] 1 and 2 only

[B] 2 and 3 only

[C] 1 and 3 only

[D] 1, 2 and 3

PYQs: Q7

Answer- D

- System of Rice Intensification (SRI) includes a method called Alternate Wetting and Drying (AWD) which is a form of controlled or intermittent irrigation of the rice crops.
- SRI methods have the following benefits and impacts, in general, compared to conventional methods of paddy cultivation:
  - To increase paddy yields usually by 20-50% and sometimes up to 100%
  - It reduces required seeds for transplanting by 60-80%(Hence Option 1 is correct);

PYQs: Q7

- The electricity consumption will be less as it enables reduce use of chemical fertilizers and agrichemicals and reduced use of irrigation water by 25-50% **(Hence Option 3 is correct)**

- **Option 2 is correct.** By stopping flooding and relying mostly on organic fertilization, methane emissions are greatly reduced without offsetting increases in nitrous oxide. Also, less carbon dioxide is generated due to reduced reliance on and transport of manufactured inputs. Oxford researchers in India found 30% reduction in net greenhouse gas emissions with SRI, compared with current practice, and more than 50% reduction in greenhouse gas emissions per kilogram of rice produced.

PYQs: Q8

2012

To meet its rapidly growing energy demand, some opine that India should pursue research and development on Thorium as the future fuel of nuclear energy. In this context, what advantage does Thorium hold over uranium?

1. Thorium is far more abundant in nature than uranium
2. On the basis of per unit mass of mined mineral, thorium can generate more energy compared to natural uranium.
3. Thorium produces less harmful waste in comparison to uranium.

Which of the statements given above is/are Correct?

a) 1 Only
b) 2 & 3 Only
c) 1 & 3 Only
d) 1,2 & 3

PYQs: Q8

Answer- D

- Thorium is 3 times as abundant as uranium
- It is estimated that 1 tonne of thorium can produce as much energy as 35 tons of uranium
- Thorium reactors are more efficient than uranium reactors, generating less waste.

PYQs: Q9

2022

With reference to India, consider the following statements:

1. Monazite is a source of rare earths.
2. Monazite contains thorium.
3. Monazite occurs naturally in the entire Indian coastal sands in India.
4. In India, Government bodies only can process or export monazite.

Which of the statements given above are correct?

a) 1, 2 and 3 only
b) 1, 2 and 4 only
c) 3 and 4 only
d) 1, 2, 3 and 4

PYQs: Q9

Answer: B

- **Statement 1 and 2 are correct.**
  - Monazite is a major source of rare earths and thorium.
  - It yields a number of rare-earth elements, such as neodymium and praseodymium. Both of these are in demand internationally for making high-performance rare-earth magnets (components of power wind turbines, electric vehicles and robotics)

PYQs: Q9

- **Statement 3 is incorrect.** Though monazite is found in most coastal areas of India. It is questionable to be found along entire coast. The main mines are found along the coasts of southern India in Kerala, Tamil Nadu and in Orissa.

- **Statement 4 is correct.** Export of monazite without a license from AERB is a violation of the Atomic Energy (Radiation Protection) Rules 2004. Indian Rare Earths Limited (IREL), a wholly owned Public Sector Undertaking of the Government of India (GOI) under DAE, is the only entity which has been permitted to produce and process monazite, and handle it for domestic use as well as for export.

PYQs: Q10

2016

India is an important member of the 'International Thermonuclear Experimental Reactor'. If this experiment succeeds, what is the immediate advantage for India?

(a) It can use thorium in place of uranium for power generation
(b) It can attain a global role in satellite navigation
(c) It can drastically improve the efficiency of its fission reactors in power generation
(d) It can build fusion reactors for power generation

PYQs: Q10

Answer- D)

The ITER (International Thermonuclear Experimental Reactor) project aims to make the long-awaited transition from experimental studies of plasma physics to full-scale electricity-producing fusion power stations.

Indian Agencies involved in it are –

- Bhabha Atomic Research Centre
- National Plasma Research Centre, Pune.

PYQs: Q11

2022

With reference to the United Nations Convention on the Law of Sea, consider the following statements:

1. A coastal state has the right to establish the breadth of its territorial sea up to a limit not exceeding 12 nautical miles, measured from baseline determined in accordance with the convention.
2. Ships of all states, whether coastal or land-locked, enjoy the right of innocent passage through the territorial sea.
3. The Exclusive Economic Zone shall not extend beyond 200 nautical miles from the baseline from which the breadth of the territorial sea is measured.

Which of the statements given above are correct?

a) 1 and 2 only
b) 2 and 3 only
c) 1 and 3 only
d) 1, 2 and 3

PYQs: Q11

Answer: D

- **Statement 1 is correct:** According to Article 3 under section 2 of United nations conventions (UNCLOS) regarding the territorial limits of the sea- every state has the right to establish the breadth of its territorial sea up to a limit not exceeding 12 nautical miles, measured from baselines determined in accordance with this Convention.
- **Statement 2 is correct:** According to Article 17 under section 3 of United nations conventions (UNCLOS) ships of all States, whether coastal or landlocked, enjoy the right of innocent passage through the territorial sea.
- **Statement 3 is correct:** The 1982 United Nations Convention on the Law of the Sea (UNCLOS) defines an Exclusive Economic Zone(EEZ). It is an area that shall not extend beyond 200 nautical miles from the baselines. The EEZ is measured from the breadth of the territorial sea.